

Planets in the Solar System: 8

Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune

Minor Celestial Objects

- **Comet** – icy object with a glowing coma and tail when near the Sun
- **Asteroid** – rocky body, mostly found in the asteroid belt between Mars and Jupiter
- **Meteoroid** – small rock or particle in space; becomes a **meteor** when it enters Earth's atmosphere, and a **meteorite** if it hits the ground

Phases of the Moon: 8

- New Moon 
- Waxing Crescent 
- First Quarter 
- Waxing Gibbous 
- Full Moon 
- Waning Gibbous 
- Last (Third) Quarter 
- Waning Crescent 

The Sun

- A **G2-type main sequence star** (yellow dwarf)
- Average surface temperature: 5,500°C
- Source of energy through **nuclear fusion** (hydrogen into helium)

Philippine Time Zone

- **GMT +8 / UTC +8** (Philippine Standard Time)

Key People in Space and Astronomy

- **Galileo Galilei** – First to use a telescope for astronomical observation; supported heliocentrism
- **Yuri Gagarin** – First human in space (April 12, 1961, Vostok 1, USSR)
- **Neil Armstrong** – First person to walk on the Moon (July 20, 1969, Apollo 11 mission, 3:56 AM UTC)
- **Laika** – First animal to orbit Earth (a dog, aboard Sputnik 2, 1957)

Largest Moon in the Solar System

- **Ganymede** (moon of Jupiter) – larger than Mercury

Largest Moon in the Solar System

- **Latin Names:**
 - **Luna** – Moon
 - **Gaia** – Earth (Greek origin, but used in astronomy)
 - **Sol** – Sun

Prograde and Retrograde Rotation

- **Prograde Rotation** – Spins in the same direction as it orbits (like Earth).

- **Retrograde Rotation** – Spins in the opposite direction of its orbit (like Venus).

Basic Facts About Earth

- **Oceans:** 5
 - Pacific, Atlantic, Indian, Southern, Arctic
- **Continents:** 7
 - Asia, Africa, North America, South America, Antarctica, Europe, Australia
 - **Pangaea** – one big supercontinent (Greek: pan = **all**, gaia = **Earth**)
 - **Panthalassa** – one big ocean (Greek: pan = **all**, thalassa = **sea**)
- **Water on Earth:**
 - Covers **71%** of the planet's surface
 - **97%** is **saltwater** (mostly oceans)
 - **3%** is **freshwater**, but most of it is locked in glaciers and ice caps

Philippine Geography

- **Provinces:** 82 — 38 in Luzon, 27 in the Visayas, and 17 in Mindanao.
- **Regions:** 17 (excluding the **Negros Island Region [NIR]**, which was dissolved)
- **Highest Peak:**
 - **Mt. Apo** (Davao Region) – 2,954 meters above sea level
- **Volcanoes:**
 - **Mt. Pinatubo** – erupted in 1991, magnitude 7.8 earthquake triggered the eruption
 - **Mayon Volcano** – known for its perfect cone shape, still active
- **Lakes:**
 - **Laguna de Bay** – largest lake in the Philippines
 - **Taal Lake** – home to **Maliputo fish**, with a volcano-island within the lake

Extreme Places

- **Hottest Place on Earth:**
 - **Death Valley, California, USA** – highest recorded temperature: 56.7°C (1913)
- **Tallest Mountain in the Solar System:**
 - **Olympus Mons** (Mars) – ~21.9 km tall, nearly 3x the height of Mt. Everest

Seasons & Climates

- **Four Seasons (Temperate Zones):**
 - **Winter** – starts in **December**
 - **Spring** – starts in **March**
 - **Summer** – starts in **June**
 - **Autumn (Fall)** – starts in **September**
- **Philippine Climate** (Tropical, with two main seasons):
 - **Dry Season: December to May**
 - Split into: **Cool Dry** (Dec–Feb), **Hot Dry** (Mar–May)

- **Wet Season: June to November**
 - Characterized by heavy rainfall and typhoons

Philippine Fauna & Floras

- **Philippine Eagle** – National bird, critically endangered
- **Sampaguita** – National flower
- **Tubbataha Reef** – UNESCO World Heritage Site

Earth's Layers (Trosmite)

- **Troposphere** – Hot air balloon, planes
- **Stratosphere** – Ozone layer, weather balloons
- **Mesosphere** – Coldest
- **Thermosphere** – Hottest
- **Exosphere** – Satellites, space shuttles

Biological Classification

6 Kingdoms of Life

- **Animalia** – multicellular, eats food (e.g., dog, cat)
- **Plantae** – makes own food, has green parts (e.g., tree, grass)
- **Fungi** – breaks down dead stuff (e.g., mushroom, mold)
- **Protista** – mostly tiny and lives in water (e.g., amoeba, algae)
- **Archaeobacteria** – lives in extreme places like hot springs (e.g., heat-loving bacteria)
- **Eubacteria** – common bacteria (e.g., bacteria in yogurt, germs)

8 Taxonomic Levels

- **Domain** – biggest group (e.g., Eukarya – has cells with nucleus)
- **Kingdom** – general group (e.g., Animalia – animals)
- **Phylum** – based on body parts (e.g., **Chordata** – has backbone)
- **Class** – more specific (e.g., Mammalia – has hair, gives milk)
- **Order** – smaller group (e.g., Carnivora – meat eaters)
- **Family** – related animals (e.g., Canidae – dog family)
- **Genus** – very close relatives (e.g., Canis – dogs and wolves)
- **Species** – exact animal (e.g., Canis lupus – wolf)

Genetics & DNA

- **Chromosomes:** 46 total
 - **44 autosomes, 2 sex chromosomes** (XX = Female, XY = Male)
- **DNA Base Pairing:**
 - Adenine (A) ↔ Thymine (T)
 - Guanine (G) ↔ Cytosine (C)
- **Trisomy 21:** Extra 21st chromosome, causes **Down Syndrome**
- **Central Dogma of Molecular Biology:**

- **Transcription:** DNA → RNA
- **Translation:** RNA → Protein

Biomolecules

- **Carbohydrates & Lipids** contain: **Carbon (C), Hydrogen (H), Oxygen (O)**
- **Cellular Respiration (Citric Acid Cycle / Krebs Cycle)** produces:
 - **Carbon Dioxide (CO₂)**
 - **ATP (energy)**

Brain Waves

- **Gamma** – Mental processing, peak cognition
- **Beta** – Alert, focused thinking
- **Alpha** – Calm, restful state
- **Theta** – Deep relaxation, light sleep, dreaming
- **Delta** – Deep sleep

Blood & Circulatory System

- **Blood Volume:** ~5 liters
- **Normal Blood Pressure:** 120/80 mmHg
- **Normal Blood pH:** ~7.4
- **Heart Chambers:** 4 (2 atria, 2 ventricles)
- **Aorta:** Largest artery
- **Capillaries:** Connect arteries and veins
- **Pericardium:** Protective sac around the heart
- **Blood Components:**
 - **Plasma** – ~55%, carries nutrients/waste
 - **Platelets** – Clotting, prevent bleeding
 - **Hemoglobin** – Carries oxygen
 - **Red Blood Cells (RBC)** – Carry CO₂ for exhalation

Comparative Hearts

- **Humans:** 1 heart
- **Earthworm:** 5 pseudohearts
- **Octopus:** 3 hearts

Human Anatomy

- **Bones:**
 - Adults: 206
 - Infants: 300
- **Muscles:** 639
- **Teeth:** 32 (adult)
- **Ribs:** 24 (12 pairs)
- **Vertebrae:** 33
- **Skull Bones:** 22
- **Facial Bones:** 14
- **Largest External Organ:** Skin
- **Largest Internal Organ:** Liver
- **Longest Bone:** Femur (thigh bone)
- **Smallest Bone:** Stapes (in the middle ear)
- **Tailbone:** Coccyx
- **Largest Muscle:** Gluteus Maximus
- **Heart Muscle Type:** Cardiac

Respiratory System

- **Nasal Cavity** – Warms, moistens air; resonance
- **Pharynx** – Passageway for air and food
- **Larynx** – Voice box
- **Trachea** – Windpipe
- **Alveoli** – Site of gas exchange

Digestive System

- **Pancreas** – Produces **insulin** and digestive enzymes
- **Liver** – Produces **bile**
- **Chyme** – Semi-liquid mixture of food from the stomach
- **Peristalsis** – Muscular movement that propels food, bile, and urine

Other Quick Facts

- **Gemstones:** Average mass 200 mg
- **Primate Classification:**
 - **Ape** – No tail
 - **Monkey** – Has tail
- **Scientific Names of Animals:**
 - Horse – *Equus*
 - Dog – *Canis*
 - Cat – *Felis*
 - Bear – *Ursus*

Evolution

- **Divergent Evolution** – Same ancestor, different traits (e.g., wolf → fox, dog)
- **Convergent Evolution** – Different ancestors, similar traits (e.g., bat & bird wings)

Plant Tropisms & Vascular Tissues

- **Positive Phototropism:** Growth toward light (usually the sunny side)
- **Positive Geotropism:** Roots grow downward toward gravity
- **Vascular Tissues:**
 - **Xylem:** Transports water and minerals
 - **Phloem:** Transports food (nutrients)

Mitosis (Cell Division)

Prophase → Metaphase → Anaphase → Telophase (PMAT)

Tissues and Structures

- **Bones** – Framework
- **Ligaments** – Bone to bone
- **Tendons** – Muscle to bone

Cell Types

- **Prokaryotic** – Unicellular (Bacteria)
- **Eukaryotic** – Multicellular (Plants, Animals, Protists, Fungi)

Tonicity

- **Isotonic** – Safe
- **Hypertonic** – Shrinks
- **Hypotonic** – Swells

Biotic & Abiotic

- **Biotic:** Plants, animals
- **Abiotic:** Water, sunlight

Circulatory System

- **Veins** – Blood back to heart
- **Arteries** – Away from heart
- **AB+** – Universal recipient
- **O-** – Universal donor

Viruses

- **Capsid** – Protein shell

Basic Ecology and Biology Terms

- **Biome** – A large region with specific climate, plants, and animals
- **Ecosystem** – Interaction of living (biotic) and non-living (abiotic) things in an area
- **Community** – Different populations of various species interacting in the same area
- **Population** – Group of organisms of the same species living in an area
- **Organism** – A single living thing

Bioremediation Methods

- **Phyto** – Using plants to clean pollutants
- **Bacterio** – Using bacteria for pollution cleanup
- **Myco** – Using fungi for bioremediation

Plant Structures and Functions

- **Breathing Organs of Plants:**
 - **Stomata** – Small pores on leaves for gas exchange
 - **Lenticels** – Small openings on trunks for gas exchange
 - **Root Hairs** – Absorb water and minerals underground
- **Monocot** Examples: Pineapple, Sugar Cane, Rice, Cassava
- **Dicot** Examples: Apple, Rose

Cell Biology

- **Cell** – Basic unit of structure and function in living things
- **Cell Membrane** – Thin structure that controls what enters and exits the cell
- **Nucleus** – Control center of the cell
- **Cytoplasm** – Jelly-like material inside the cell holding organelles
- **Cilia** – Hair-like structures for movement or sensing
- **Flagella** – Whip-like tail for movement
- **Vacuole** – Storage area for water and nutrients
- **Lysosomes** – Break down waste materials
- **Ribosomes** – Site of protein synthesis

Cellular Processes

- **Cellular Respiration:** Process by which cells convert oxygen and glucose into energy (ATP), carbon dioxide, and water
- **Energy (ATP)** – Main energy currency of the cell

- **ION (In Oxygen Night)** – Refers to oxygen use in respiration

Fungal Cell Wall Components

- **Glucans/Sugar**
- **Chitin**
- **Glycoproteins**

Symbiotic Relationships

- **Mutualism** – Both species benefit
 - Example: Anemones & Clownfish, Corals & Algae
- **Commensalism** – One benefits, the other is unaffected
 - Example: Whale & Barnacle
- **Parasitism** – One benefits at the expense of the other
 - Example: Dog & Tick

Ecological Interactions

- **Predation** – One organism eats another
 - Example: Lion & Deer
- **Grazing** – Herbivores feeding on plants
 - Example: Carabao & Grass
- **Competition** – Organisms competing for the same resources
 - Example: Mango Tree & Sampaguita

Human Anatomy and Physiology Terms

- **Esophagus** – Food pipe
- **Nearest to Heart** – (Context unclear, possibly referring to esophagus or other organ)
- **Larynx** – Voice box
- **Lungs** – Main organ for respiration
- **Liver** – Secretes bile for digestion
- **Pancreas** – Produces insulin
- **Pancreatic Cells (Islets of Langerhans)** – Cells that produce insulin

Newton's Laws of Motion

1. **Law of Inertia** – A body remains at rest or in uniform motion unless acted upon by a force.
2. **Law of Acceleration** – More force means more acceleration.
3. **Law of Interaction** – For every action, there's an equal and opposite reaction.

Types of Forces

- **Friction** – Opposes motion.
- **Gravity** – Pulls objects toward Earth.
- **Tension** – Force through strings/ropes.
- **Normal Force** – Support force from surfaces.

Concepts

- Heavier objects require more force to move.
- Friction slows things down, but also allows walking and driving.

Heat and Thermodynamics

Heat – Form of energy that moves from hot to cold objects.

Temperature – Measure of how hot/cold an object is.

Heat Transfer Methods

1. **Conduction** – Through solids (e.g., metal spoon gets hot).
2. **Convection** – Through fluids (e.g., boiling water).
3. **Radiation** – Through space (e.g., sunlight).

Laws of Thermodynamics

- Energy cannot be created or destroyed, only transferred.
- Heat flows naturally from hot to cold.

Types of Light Behavior

- **Reflection** – Light bounces (mirror).
- **Refraction** – Light bends (straw in water looks bent).
- **Diffraction** – Light spreads around corners.
- **Absorption** – Light is taken in by material.

Lenses

- **Convex** – Converges light, used in magnifying glasses.
- **Concave** – Diverges light, used in eyeglasses for nearsightedness.

Current Electricity

- **Current** – Flow of electric charge.
- **Voltage** – Electric pressure (pushes current).
- **Resistance** – Slows down current (depends on material).

Circuits

- **Series** – One path only; if one bulb goes out, all do.
- **Parallel** – Multiple paths; one bulb can go out, others stay on.

Magnetism

- Magnetic fields are strongest at the poles.
- Electricity can produce magnetism (electromagnet).
- Earth acts like a giant magnet.

Types of Magnetism

- **Diamagnetic** – Weakly repelled (e.g., gold)
- **Paramagnetic** – Weakly attracted (e.g., aluminum)
- **Ferromagnetic** – Strongly attracted (e.g., iron, nickel, cobalt)

Wave Basics

- **Mechanical waves** – Need a medium (e.g., sound).
- **Electromagnetic waves** – Don't need a medium (e.g., light).

Sounds

- Travels faster in solids, slower in gases.
- Cannot travel in a vacuum (no air, no sound).
- **Pitch** = High or low sound (related to frequency).
- **Loudness** = Volume (related to amplitude).

Wave Properties

- **Frequency** – How many waves per second.
- **Amplitude** – Height of the wave (energy).
- **Wavelength** – Distance between waves.

Free Fall

- All objects fall at the same rate (ignoring air resistance).
- Gravity pulls at approx. **9.8 m/s²**.

Projectile Motion

- Curved path of an object thrown into the air.
- Horizontal and vertical motions are independent.

Measurement & Tools in Physics

- **Thermometer** – Measures temperature.
- **Barometer** – Measures air pressure.
- **Sphygmomanometer** – Measures blood pressure.
- **Stethoscope** – Listens to body sounds.
- **Seismograph** – Detects earthquakes.
- **Multimeter** – Measures voltage, current, and resistance.

Basic SI Units

- Length – meter (m)
- Mass – kilogram (kg)
- Time – second (s)
- Temperature – kelvin (K)
- Electric current – ampere (A)

Magnetism

- **Diamagnetic** – Gold (Au)
- **Paramagnetic** – Aluminum (Al)
- **Ferromagnetic** – Iron (Fe), Nickel (Ni), Cobalt (Co)

Basic Atomic Facts

Total Elements: 118

- **Atomic Structure:**
 - Protons = Atomic Number
 - Electrons = Protons (in neutral atom)
 - Neutrons = Mass Number – Atomic Number

Periodic Table

- **Rows (Horizontal): 7 Periods**
- **Columns (Vertical): 18 Groups/Families**
- **Blocks:**
 - **s-block:** Groups 1–2 (2 elements wide)
 - **p-block:** Groups 13–18 (6 elements wide)

- **d-block:** Transition metals (10 elements wide)
- **f-block:** Lanthanides & Actinides (14 elements wide)

- **Mendeleev:** Father of the Periodic Table (left gaps for undiscovered elements)
- **Moseley:** Arranged the table by **atomic number** (modern version)

Gases and the Atmosphere

Most Abundant Gases in Earth's Atmosphere:

- **Nitrogen (N₂)** – ~78%
- **Oxygen (O₂)** – ~21%
- **Argon (Ar)** – ~0.93%
- **Carbon Dioxide (CO₂)** – ~0.04% (but major greenhouse gas)

Note:

The CO₂ and CO values given earlier (e.g., 82%, 49.1%) refer to their contribution to **pollutants or greenhouse effects**, not atmospheric composition.

- **CO₂** – Major **greenhouse gas**
- **CO** – **Carbon monoxide**, a toxic pollutant from incomplete combustion
- **Ozone (O₃)** – Found in the stratosphere, absorbs UV radiation

Key Element Properties & Facts

- **H₂O** – **Dihydrogen monoxide** (Water)
- **NaCl** – **Sodium chloride** (Table salt)
- **Cu (Copper)** – Best metal conductor of electricity
- **Ag (Silver)** – Best overall conductor (but expensive)

Flame Test Colors (Common Lab Use)

- **Boron (B)** – Bright green
- **Iron (Fe)** – Golden yellow
- **Calcium (Ca)** – Orange-red
- **Sodium (Na)** – Bright yellow
- **Potassium (K)** – Lilac/violet
- **Copper (Cu)** – Blue-green
- **Strontium (Sr)** – Bright red
- **Lithium (Li)** – Crimson
- **Barium (Ba)** – Pale green

Notable Scientists

- **Dmitri Mendeleev** – Father of the Periodic Table
- **Henry Moseley** – Arranged periodic table by atomic number
- **Joseph Priestley** – Discovered **Oxygen** (1774), called it "dephlogisticated air"
- **Marie Curie** – Discovered **Radium** and **Polonium**, pioneer of radioactivity
- **Antoine Lavoisier** – Father of Modern Chemistry; named Oxygen & Hydrogen; Law of Conservation of Mass

Common Compounds & Ions

- **H₂O** – Water
- **NaCl** – Table salt
- **CO₂** – Carbon dioxide
- **NH₃** – Ammonia

- **CH₄** – Methane
- **HCl** – Hydrochloric acid
- **H₂SO₄** – Sulfuric acid (battery acid)
- **NaOH** – Sodium hydroxide (lye)

Periodic Table Groups Quick Recap

- **Group 1** – Alkali metals (very reactive, soft)
- **Group 2** – Alkaline earth metals
- **Groups 3–12** – Transition metals
- **Group 17** – Halogens (very reactive nonmetals)
- **Group 18** – Noble gases (inert, stable)

Chemical Safety/Handling Tips

- Always label chemical containers
- Wear gloves, goggles, lab coat
- Acids → Always add **acid to water**, not the reverse
- Dispose of chemicals properly per lab protocol

Philippine History

- **Oldest City & First Capital:** Cebu
- **St. Augustine** – Faith
- **St. Thomas Aquinas** – Reason

Largest Provinces (PA-ISA-LAB)

1. Palawan
2. Isabela
3. Lanao del Norte

UNESCO World Heritage Sites (TPH-BCV)

- Tubbataha Reef
- Puerto Princesa Subterranean River
- Hamiguitan Range
- Baroque Churches
- Cordillera Rice Terraces (Ifugao)
- Vigan

5 Pillars Of Islam

1. **Shahada** – Faith
2. **Salah** – Prayer
3. **Sawm** – Fasting
4. **Zakat** – Almsgiving
5. **Hajj** – Pilgrimage to Mecca

Holy Books & Religions

- **Qur'an** – Islam
- **Bible** – Christianity
- **Torah** – Judaism
- **Bhagavad Gita** – Hinduism

Missionary Orders (AFJDR-B)

1. Augustinians
2. Franciscans
3. Jesuits
4. Dominicans
5. Recollects
6. Benedictines

Hinduism (TRIMURTI)

- **Brahma** – Creator (4 heads)
- **Vishnu** – Preserver

- **Shiva** – Destroyer

Other Beliefs

- **Paganism** – Polytheistic (many gods)
- **Animism** – Nature spirits, mediated by **babaylan**
- **Bathala** – Supreme being

**UN SECURITY COUNCIL
Permanent 5 (CFRUU)**

- China
- France
- Russia
- United Kingdom
- United States

Philippine Revolution

- **Martial Law** – August 30, 1896
- **Governor-General:** Ramón Blanco
- **8 Provinces:** Manila, Cavite, Batangas, Laguna, Tarlac, Nueva Ecija, Bulacan, Pampanga

Filipino Deities

- **Apo Laki** – Fertility and war
- **Ideanale** – Labor and good deeds

Human Act and Act of Man

- **Human Act** – Intentional (e.g., self-care)
- **Act of Man** – Unintentional (e.g., breathing)

Asean Member States (10)

1. Brunei
2. Cambodia
3. Indonesia
4. Laos
5. Malaysia
6. Myanmar
7. Philippines
8. Singapore
9. Thailand
10. Vietnam

Mga Bantas (Punctuation Marks)

1. Panipi (“ ”) — Quotation Marks

Ginagamit sa direktang sinasabi ng isang tao o kapag binabanggit ang pamagat ng tula, kanta, kwento, o artikulo.

Halimbawa:

Sabi niya, “Mahalaga ang edukasyon.”
Binasa ko ang “Ibong Adarna.”

2. Panaklong () — Parentheses

Ginagamit sa dagdag na paliwanag o impormasyon tulad ng taon, kahulugan, o posisyon.

Halimbawa:

Si Jose Rizal (bayani ng Pilipinas) ay mahusay magsulat.

Ang WHO (World Health Organization) ay tungkol sa kalusugan.

3. Kuwit (,) — Comma

Ginagamit para paghiwalayin ang mga salita sa serye o magpahinto ng bahagya sa pangungusap.

Halimbawa:

Bumili siya ng papel, lapis, at pambura.
Oo, pupunta ako.

4. Tuldok (.) — Period

Sa dulo ng pangungusap na nagsasalaysay o nag-uutos.

Halimbawa:

Umuulan ngayon.
Mag-aral ka.

5. Tandang Pananong (?) — Question Mark

Sa tanong o pangungusap na nagtatanong.

Halimbawa:

Anong oras na?
Kumain ka na ba?

6. Tandang Padamdang (!) — Exclamation Point

Ginagamit kapag nagpapakita ng matinding damdamin tulad ng galit, tuwa, takot, o gulat.

Halimbawa:

Ang ganda mo!
Aray!

7. Gitling (–) — Hyphen

Ginagamit kapag inuulit ang salita o kapag may panlaping sinusundan ng salitang-ugat na nagsisimula sa patinig.

Halimbawa:

Araw-araw tayong nagkikita.
Mag-aral ka.

8. Tutuldok (:) — Colon

Ginagamit bago magbigay ng halimbawa, paliwanag, o direktang sipi.

Halimbawa:

Kailangan ko ng: papel, lapis, at notebook.
Sabi ng guro: "Basahin ang kabanata."

9. Tutuldok-kuwit (;) — Semicolon

Ginagamit kung may dalawang ideya sa iisang pangungusap na magkaugnay pero buo ang diwa.

Halimbawa:

Mahilig siyang kumanta; mahilig din siyang sumayaw.

10. Kudlit (') — Apostrophe

Ginagamit sa pinaikling salita o informal na usapan.

Halimbawa:

'Di ko alam. (Hindi ko alam.)
Kumain 'ko n'un. (Kumain ako noon.)

PRESIDENTS OF THE PHILIPPINES

1st Republic (Revolutionary Government)

Emilio Aguinaldo (1899–1901)

- *El Presidente*
- First President of the Philippines
- Declared Philippine Independence (June 12, 1898)
- Introduced the Philippine flag
- Youngest president (29 years old)
- Revolutionary Government Leader
- Known as the **Undisputed President**

Commonwealth of the Philippines (1935–1946)

Manuel L. Quezon (1935–1944)

- *Ama ng Wikang Pambansa*
- Declared **Tagalog** as the national language
- Advocated **women's suffrage**
- Established the **National Economic Council**
- *Father of Social Justice*
- Created **National Land Settlement Administration (NLSA)**
- Quote: "My loyalty to my party ends where my loyalty to my country begins."

Sergio Osmeña (1944–1946)

- Rehabilitated **Philippine National Bank (PNB)**
- Helped PH join **IMF**
- 1st **Visayan** president
- **Shortest-serving** president
- **First Speaker** of the **Philippine Assembly**

2nd Republic (Japanese Occupation)

Jose P. Laurel (1943–1945)

- Initiated "**Magtanim ay Di Biro**" program
- President under **Japanese-sponsored Republic**
- Branded as **Puppet President**
- Founder of **Lyceum of the Philippines**
- Served all **3 branches of government**:
 - Senator
 - Congressman
 - Associate Justice

3rd Republic (Post-WWII Independence)

Manuel Roxas (1946–1948)

- Nickname: *Manoling*
- Signed **Philippine Rehabilitation Act**
- Granted **amnesty** to WWII collaborators
- Implemented the **70-30 policy** favoring Filipinos
- Began economic and political recovery

Elpidio Quirino (1948–1953)

- Created **Integrity Board** to combat corruption
- Established factories, built a **hydroelectric plant**
- First **Ilocano** president
- Founded **LASEDECO** (Land Settlement and Dev't Corporation)

- Strengthened **Social Security System (SSS)**
- Known for the *Golden Arinola story*

Ramon Magsaysay (1953–1957)

- Nicknames: *Man of the Masses*, *Little Man*
- Investigated and eliminated corruption
- Implemented **NARRA** (National Resettlement and Rehabilitation Administration)
- Created **PCAC** (Presidential Complaints and Action Committee)
- Suppressed the **Hukbalahap Movement**
- Promoted **SEATO** membership
- Signed **Rizal Law**
- Opened **Malacañang** to the public
- Quote: *“Those who have less in life should have more in law.”*

Carlos P. Garcia (1957–1961)

- Promoted **Filipino First Policy**
- Launched **Austerity Program**
- Signed **Bohlen-Serrano Agreement**
- *Prince of Visayan Poets / Bard from Bohol*
- First president buried in **Libingan ng mga Bayani**

Diosdado Macapagal (1961–1965)

- Nickname: *Poor Boy from Lubao*
- Enacted the **Agricultural Land Reform Code**
- Passed **Minimum Wage Law**
- Changed **Independence Day** to **June 12**
- Promoted land ownership (up to **75 hectares**)
- Created **Veterans Bank** and **Land Bank**
- Quote: *“I have sat at the sumptuous tables of power, but I have never lost my sense of humility.”*

4th Republic (Martial Law Period)

Ferdinand E. Marcos Sr. (1965–1986)

- Nickname: *Macoy*
- Promoted **Miracle Rice**, built health centers, schools, roads
- Reformed **justice system** and **AFP**
- Declared **Martial Law** under *Proclamation 1081*
- Transitioned to **Parliamentary Government**
- Ousted via **EDSA People Power Revolution** (Feb. 25, 1986)
- Quote: *“I will make this country great again.”*

5th Republic (Post-EDSA 1986–present)

Corazon Aquino (1986–1992)

- *People Power President*, 1st Woman President in Asia
- Known as *Mother of Democracy*
- Promulgated **1987 Constitution** (Freedom Constitution)
- Introduced **Family Code of 1987**

- Enacted **CAR Law (RA 6657)** → continued by **CARP (RA 9700)**
- Reformed **Executive** and **Supreme Court**

Fidel V. Ramos (1992–1998)

- Solved **Power Crisis of the 1990s**
- Signed peace with **MNLF**
- Launched **Philippines 2000** vision
- Mitigated **1997 Asian Financial Crisis**
- Passed **IPRA (RA 8371)** – Indigenous Peoples’ Rights Act

Joseph Estrada (1998–2001)

- Nickname: *Erap*, *Ama ng Masa*
- Created **National Anti-Poverty Commission**
- Ordered **MILF** camp seizure
- Launched **Enterprise Dev’t for the Poor**
- Formed **Presidential Commission on Higher Education Reform**
- Ousted via **EDSA II** (2001)
- Quote: *“Erap Para sa Mahirap”*

Gloria Macapagal Arroyo (2001–2010)

- Took oath on **Jan. 20, 2001** (EDSA II)
- Advocated **Holiday Economics**
- Passed **EVAT (Expanded VAT)**
- Promoted **Strong Republic** slogan
- Known for **Hello Garci scandal**
- Quote: *“I am sorry.”*

Benigno "Noy" Aquino III (2010–2016)

- Nickname: *P-Noy*
- Enforced **No Wang-Wang Policy**
- Investigated **PDAF Scam (Pork Barrel)**
- Implemented **K–12 Education Program**
- Improved **PAGASA/DOST**
- Signed **RH Law**
- Formed **Truth Commission**
- Won **arbitral ruling vs. China**
- Quote: *“The Filipino is worth dying for.”*

Rodrigo Duterte (2016–2022)

- 1st **Mindanaoan** president
- Advocated **Federalism**
- Launched **War on Drugs**
- Passed **TRAIN Law**
- Signed **Bangsamoro Organic Law**
- Built **Malasakit Centers**
- Passed **Universal Access to Tertiary Education Act**
- Initiated **PhilSys (National ID System)**
- Launched **Build Build Build**

Current President

Ferdinand R. Marcos Jr. (2022–present)

- Son of Ferdinand Marcos Sr.
- 17th President of the Philippines
- **SIM Card Registration Act of 2022**
- **Maritime Zones Act of 2024**
- **Self-Reliant Defense Posture Act of 2024**

- Philippine Archipelagic Sea Lanes Act of 2024
- Anti-Agricultural Economic Sabotage Act of 2024
- New Government Procurement Act (RA 12009)
- E-Government and E-Governance Act (EBET Framework Act, RA 12063)
- Agricultural Tariffication Act (RA 12078)
- VAT Refund for Non-Resident Tourists Act (RA 12079)
- Basic Mental Health and Well-Being Promotion Act (RA 12080)
- Executive Order No. 6 – BARMM Reorganization
- Executive Order No. 35 – Pasig River Urban Development

Sociology

- **Gesellschaft** – Society (transaction-based)
- **Gemeinschaft** – Community (shared experience)

Filipino / Linguistics

- **Baybayin**: 17 symbols
- **ABAKADA**: 20 letters
- **English Alphabet**: 26
- **Binagong Alpabeto**: 28
- **Pinagyamang Alpabeto**: 31
- **Ponema**: 21 meaningful sounds
- **Terms**:
 - **Ponolohiya**: Study of sounds
 - **Morpema**: Smallest meaningful unit
 - **Grapema**: Symbol representing sound
 - **Digrapo**: 2 letters, 1 sound (e.g., NG)
 - **Alomorp**: Variations of a morpheme
 - **Alopono**: Sound variations
 - **Tambalan**: Compound words (e.g., bahay-kubo, anak-araw)

Figures Of Speech (Tayutay)

1. **Pagtutulad** – Simile
2. **Pagwawangis** – Metaphor
3. **Pagbibigay-katauhan** – Personification
4. **Pagmamalabis** – Hyperbole
5. **Apostrophe** – Addressing absent/dead
6. **Pagpalit-tawag** – Metonymy
7. **Pagpapalit-saklaw** – Synecdoche
8. **Paghihimig** – Onomatopoeia
9. **Pagsalungat** – Oxymoron
10. **Paralelismo** – Parallelism
11. **Paglumanay** – Euphemism

Di-verbal Na Komunikasyon

1. **Oculesics** – Eye movement
2. **Haptics** – Touch
3. **Kinesics** – Body movement
4. **Objectics** – Objects used
5. **Olfactorics** – Smell
6. **Colorics** – Color usage
7. **Pictics** – Facial expressions
8. **Iconics** – Symbols/icons

9. **Chronemics** – Use of time
10. **Vocalics** – Voice/sounds
11. **Proxemics** – Space/distance

Exploration And Circumnavigation

- **Magellan (1521)** – First expedition to circumnavigate the globe. Arrived in the Philippines, named it **Archipelago of St. Lazarus**.
- **Elcano** – First man to complete the circumnavigation after Magellan’s death.
- **Expedition Ships (TSVSC)**:
 - *Trinidad* (Magellan)
 - *San Antonio*
 - *Victoria* (Elcano)
 - *Santiago*
 - *Concepcion*
- **Sikatuna & Legazpi** – Blood Compact (Sanduguan) in Bohol.
- **Magellan & Kulambo** – Rajah of Limasawa.
- **Spanish Expeditions (MaloCa → SaViPi)**:
 - *Magellan* (1521)
 - *Loaisa*
 - *Cabot*
 - *Saavedra*
 - *Villalobos* – Named the islands *Las Islas Filipinas*
 - *Legazpi* – Established Spanish rule.

Social Classes During Spanish Colonization

- **Peninsulares** – Spaniards born in Spain.
- **Insulares** – Spaniards born in the Philippines.
- **Mestizo** – Mixed race; *dugong banyaga*.
- **Principalia & Ilustrados** – Educated middle class.
- **Indio** – Native Filipinos (e.g., Bonifacio).

Migration Theories

Beyer’s Migration Theory (DAIM):

- *Dawn Man* – Primitive human.
- *Aeta* – First settlers, aboriginal.
- *Indonesians* – First seafarers.
- *Malay* – Last to come.

Jocano’s Core Population Theory:

- Indigenous evolution, *Tabon Man* (Palawan).
- Discovered by *Fox & Santiago*.

Treaties

- **Treaty of Paris (1898)** – Philippines ceded to the US; \$20M.
- **Treaty of Tordesillas** – Divided East (Portugal) and West (Spain); by Pope Alexander VI.
- **Biak-na-Bato** – Truce.
- **Tydings-McDuffie Act** – 10-year transition to independence.

Propaganda Movement

- Led by **Ilustrados**.
- **Triumvirate**:
 - *Rizal* – *Noli Me Tangere*, *El Filibusterismo*

- **Graciano Lopez Jaena** – *Fray Botod* (Diego Laura)
- **Marcelo H. del Pilar** – *Frailocracy, Dasalan at Tocsohan*

Early Trade & Culture

- **First Traders:**
 - *Orang Dampuan* – From Champa (Vietnam)
 - *Banjarmasin* – Indonesia
 - *Chinese* – Ming Dynasty era
- **Writing Systems:**
 - *Cuneiform* – Sumerians
 - *Hieroglyphics* – Egyptians
 - *Indus Script* – India
 - *Baybayin/Alibata* – Philippines
 - *Hangul* – Korea

Item Difficulty Index

- 0.81–1.00 – Very Easy
- 0.61–0.80 – Easy
- 0.41–0.60 – Average
- 0.21–0.40 – Difficult
- 0.00–0.20 – Very Difficult

Item Evaluation

- 0.40+ – Very Good (Retain)
- 0.30–0.39 – Good (Revise)
- 0.20–0.29 – Fair (Revise)
- 0.09–0.19 – Poor (Reject)

Ethical Sections

- I. Scope & Limitations – Applies to all levels.
- II. Teacher & the State – Avoid political bias.
- III. Teacher & the Community – Maintain good relationships.
- IV. Teacher & the Profession – Continue CPE.
- V. Loyalty to the Profession.
- VI. Teacher & Higher Authorities – Follow proper channels.
- VII. School Officials and Teachers – No corporal punishment, no favors/gifts.
- VIII. Parents – Cordial communication.
- IX. Learners – Free tutorials, avoid favoritism.
- X. Business – Allowed, not during duty hours.
- XI. Teacher as a Person – Dignified conduct.
- XII. Disciplinary Actions – RA 7836.
- XIII. Effectivity – Within 60 days.

Grading Scale

- 90–100 – Outstanding
- 85–89 – Very Satisfactory
- 80–84 – Satisfactory
- 75–79 – Fairly Satisfactory
- Below 75 – Did Not Meet Expectations

Assessment AS, FOR, and OF Learning

1. Assessment AS Learning

- Students monitor and reflect on their own learning.
- They become aware of their progress and set goals.
- Example: self-assessment, learning journals.

2. Assessment FOR Learning

- Ongoing checks to guide teaching and help students improve.
- Formative assessments used to give feedback during the learning process.
- Example: quizzes, class discussions, draft assignments.

3. Assessment OF Learning

- Measures what students have learned at the end of a period.
- Summative assessments used for grading or reporting.
- Example: final exams, standardized tests, end-of-unit projects.

Types Of Assessment

1. Formative Assessment

- Done during learning to monitor progress.
- Helps improve learning.
- Examples: quizzes, observations, feedback.

2. Summative Assessment

- Done after learning to evaluate achievement.
- Used for final grades or decisions.
- Examples: final exams, major projects, standardized tests.

3. Diagnostic Assessment

- Done before instruction to find students' strengths and weaknesses.
- Helps teachers plan lessons.
- Examples: pre-tests, initial interviews.

4. Performance-Based Assessment

- Students show what they can do through tasks or projects.
- Focuses on real-life skills.
- Examples: presentations, experiments, portfolios.

Curriculum Types

- **Written** – Syllabi, modules, books.
- **Taught** – Implemented by teachers.
- **Supported** – Print, tech.
- **Assessed** – Evaluated.
- **Learned** – Behavioral change.
- **Hidden** – Indirect effects (peer influence, calamities).
- **Recommended** – Found in DepEd.

Curriculum Builders

- **Knower** – Subject mastery.
- **Writer** – Creates content.
- **Planner** – Plans activities.
- **Innovator** – Creative approach.
- **Implementer** – Executes curriculum.
- **Evaluator** – Monitors outcomes.

Educational Philosophies

1. Traditional/Conventional (Teacher-Centered):

- *Hutchins* – Perennialism.
- *Bestor* – Essentialism.
- *Schwab* – Discipline.
- *Phenix* – Cognitive.

2. Progressive (Learner-Centered):

- *Dewey* – Progressivism.
- *Caswell & Campbell* – Teacher guidance.
- *Smith, Stanley, Shore* – Sequence in schools.
- *Marsh & Wills* – Teachers plan/enact.

Curriculum Models & Theories

Kurt Lewin's Force Field (SAREPV):

- Substitution, Alteration, Restructuring, etc.

Ralph Tyler (Deductive):

1. Objectives
2. Select Activities
3. Organize
4. Evaluate

Hilda Taba (Inductive):

1. Diagnose
2. Objectives
3. Content Selection/Organization
4. Activities
5. Evaluation

Industrial Revolutions

- 1.0 – Steam
- 2.0 – Electricity
- 3.0 – Digital
- 4.0 – Internet

Dale's Cone Of Experience

- 10% Read
- 20% Hear
- 30% See
- 50% Hear and See
- 70% Say and Write
- 90% Do, Demonstrate

Disabilities & Impairments

- **Impairment** – Loss/abnormality.
- **Disability** – Functional limitation.
- **Handicap** – Disadvantage.
- **Decibel** – Loudness.
- **Hertz** – Frequency.
- **Paraplegia** – Lower body paralysis.
- **Quadriplegia** – Full limb paralysis.
- **Cerebral Palsy** – Movement disorder.
- **Fibromyalgia** – Muscle/tissue pain.
- **Down Syndrome** – Extra chromosome (21st).

5S (Organization Approach)

1. **Sort** – Selecting.
2. **Set** – Organizing.
3. **Shine** – Cleaning.
4. **Standardize** – Setting rules.
5. **Sustain** – Monitoring.

Classroom Management (SLP-432)

- **Setting** – Physical layout.
- **Learner** – Focused on experience.
- **Problem** – Real-life scenarios.
- **Conduct Management** – Discipline.
- **Climate** – Mood/atmosphere.
- **Proactive** – Before misbehavior.
- **Reactive** – During misbehavior.
- **Punitive** – After misbehavior.

Notable Figures and Groups

- **Cecile Alvarez** – PETA
- **Naty Rogers** – SALA
- **Dolores Manapat** – *Piping Dilat*
- **Diego Laura** – *Fray Botod*
- **Marcelo H. Del Pilar** – *Plaridel*, 2nd editor of *La Solidaridad*, known as the "Great Orator"
- **Graciano Lopez Jaena** – *La Solidaridad*, *Fray Botod*

Literary Forms

- **Haiku** – 5-7-5 syllables, nature-themed, introduced by Matsuo Basho
- **Tanka** – 5-7-5-7-7 (31 syllables), themes of love and nature

Sculpture Methods (CAMC)

- Carving
- Assembling
- Modeling
- Casting

Art Elements

- **Line**
- **Texture**
- **Shape**
- **Value**
- **Form**
- **Color**
- **Space**

Significant Events and Dates

- **March 31, 1521** – First Mass (Limasawa-Valderama dispute)
- **April 7, 1521** – Magellan arrived in Cebu
- **April 21, 1521** – Baptism of Rajah Humabon and wife
- **April 27, 1521** – *Battle of Mactan*; death of Magellan

Traditional Clothing

- **Kangan** – upper garment
- **Bahag** – lower garment
- **Putong** – headgear for males

Spanish Colonial Period

Governor Generals:

- **Miguel López de Legazpi** – first
- **Diego de los Ríos** – last
- **Dela Torre** – most loved
- **Clavería** – implemented surnames
- **Basco** – tobacco monopoly

- **Izquierdo** – Cavite Mutiny
- **Despujol** – exiled Rizal to Dapitan
- **Polavieja** – signed Rizal’s execution

Policies and Systems:

- **Pueblo** – town
- **Cabecera** – town center
- **Ayuntamiento** – trade/governance center
- **Reducción** – resettlement policy
- **Tributo** – tax
- **Bandala** – mandatory sale of crops
- **Polo y Servicio** – forced labor
 - *Polista* – forced laborer
 - *Falla* – exemption fee
- **Galleon Trade** – between Philippines and Mexico
- **Alcaldia** – pacified provinces
- **Corregimientos** – unpacified areas
- **Indulto de Comercio** – trade privilege
- **Frailocracy** – friar-led governance
- **Malohokan** – town crier
- **Royal Audiencia** – Supreme Court

Pre-colonial Education And Society

- **Bothoan** – first school
- **Parents** – first teachers
- **Primitive education** – survival-based, essentialist
- **Cordilleras** – Igorot mummification
- **Pintados** – tattooed warriors from Panay

Philippine Revolutionary Period

Key Figures:

- Apolinario Mabini – "Brain of the Revolution"
- Emilio Jacinto – "Brain of the Katipunan"

Katipunan (KKK):

- Secret society aiming for independence
- **Objectives:** Political, Moral, Civil
- Trial by Ordeal – water, fire, wounds (believed to show divine intervention)
- *Tatlong Kawal (Three Soldiers):*
 1. Arellano – Anak ng Bayan
 2. Basa – Kawal, GomBurZa
 3. Bonifacio – Bayani, Rizal
- Katipunan led to Rizal’s execution
- Father Mariano Gil – priest who reported the Katipunan

Colonial And Post-colonial Transition

American Period:

- **Military Governors:** Merritt, Otis, Arthur MacArthur
- **Civil Governors:** Taft, Murphy
- Commonwealth Government

Laws and Acts:

- **Gabaldon Act** – establishment of public schools
- **Bell Trade Act** – parity rights, no tariffs, fixed exchange rate
- **Tydings-McDuffie Act** – Philippine Independence Act, 10-year transition
- **Jones Law** – introduced bicameralism
- **Cooper Act** – established civil government

International Organizations

- **NATO** – North Atlantic Treaty Organization
 - Formed during Cold War
 - Promotes democratic values
 - Opposed USSR/Soviet Union
- **ASEAN** – Association of Southeast Asian Nations
 - Promotes regional cooperation
 - Core principles: consensus, non-interference
 - Covers political, social, economic issues
- **SEATO** – Southeast Asia Treaty Organization
- **APEC** – Asia-Pacific Economic Cooperation
- **OPEC** – Organization of Petroleum Exporting Countries
 - Controls global oil supply
- **WTO** – World Trade Organization
 - Oversees international trade rules
 - Monitors tariffs and trade disputes

Famous World Landmarks

- **Eiffel Tower** – Paris, France
- **Big Ben** – London, UK
- **The Colosseum** – Rome, Italy
- **Parthenon** – Athens, Greece
- **Pyramid of Giza** – Egypt
- **Machu Picchu** – Peru
- **Angkor Wat** – Cambodia

Legal And Governmental Terms

- **Cabeza de Barangay** – tax collector
- **Anod Bayan** – Ombudsman equivalent

Famous Artists And Their Works

1. **Leonardo da Vinci**
 - *Mona Lisa, The Last Supper, Vitruvian Man, Annunciation, Lady with an Ermine*
2. **Vincent van Gogh**
 - *The Starry Night, Sunflowers, Café Terrace at Night, Irises, The Potato Eaters*
3. **Fernando Amorsolo**
 - *President Aguinaldo, The Burning of Manila, The Making of the Philippine Flag, Planting Rice, Tinikling Dance*
4. **Michelangelo**
 - *The Last Judgment, Sistine Chapel Ceiling, The Creation of Adam, The Crucifixion of St. Peter, Pietà*

Philosophical Views On Art

- **Plato** – Art as *Imitation* (Mimesis)
- **Aristotle** – Art as *Representation*
- **Immanuel Kant** – Art as *Disinterested Judgment*, “Art for Art’s Sake”
- **Leo Tolstoy** – Art as *Communication of Emotion*

Art Movements & Styles

- **Renaissance / Realism** – Da Vinci, Michelangelo

- **Baroque** – Caravaggio, Rembrandt (*Dramatic*)
- **Impressionism** – Monet, Degas (*Light & Movement*)
- **Modern / Abstract / Experimental** – Picasso, Dali
- **Contemporary** – *Socially relevant, mixed media, digital*

Philippine Visual Artists

- **Amorsolo** – Rural landscapes
- **Carlos “Botong” Francisco** – Depicted farmers and fisherfolk
- **Victorio Edades** – Father of Modern Philippine Painting
- **Benedicto Cabrera (BenCab)** – Social Realism, “Sabel”

Filipino Writers And Literary Figures

- **Graciano López Jaena** – Writer & Orator, *Founder of La Solidaridad*, championed *freedom and human rights*
- **Francisco Balagtas** – *Florante at Laura*; political allegory
- **Amado V. Hernandez** – Labor leader, poet, and novelist
- **Nick Joaquin** – *May Day Eve*, Magical Realism
- **Francisco Arcellana** – Modern short stories pioneer
- **F. Sionil Jose** – *Rosales Saga*, social justice
- **Edith Tiempo** – Love sonnets
- **Jose Garcia Villa** – “Comma Poet”
- **Juan Luna** – Painter & revolutionary
 - *Spoliarium, The Death of Cleopatra, La Batalia*
 - Member of *Propaganda Movement*
- **Antonio Luna** – Writer, musician, military strategist
 - Wrote in *La Solidaridad*
 - Played guitar and piano

Philippine Theater And Film

- **Lamberto Avellana** – *Anak Dalita*
- **Ishmael Bernal** – *Himala*
- **Lino Brocka** – *Maynila: Sa Kuko ng Liwanag*
- **Wilfrido Ma. Guerrero** – *Forever, Wanted: A Chaperone*
- **Rolando Tinio** – Experimental theater
- **Cecile Guidote-Alvarez** – PETA founder (educational & political theater)
- **Naty Rogers** – SALA
- **Alice Reyes** – Founder of Ballet Philippines
- **Gantimpala Theater** – *Florante at Laura*
- **Tanghalang Pilipino** – Cultural Center of the Philippines' resident theater company
- **Repertory Philippines** – Broadway-style productions

Philippine Music And Composers

- **Lucio San Pedro** – *Sa Ugoy ng Duyan*
- **Antonio Molina** – *Hatinggabi*
- **Levi Celerio** – “Leaf Musician,” 4000+ songs
- **Ryan Cayabyab** – *Kay Ganda ng Ating Musika*, Contemporary composer
- **Andrea Veneracion** – Founder of the Philippine Madrigal Singers

Dance Personalities

- **Francisca Reyes-Aquino** – Documented Philippine folk dances
- **Leonor Goquingco** – Mother of Philippine Theater Dance
- **Alice Reyes** – Ballet Philippines
- **Ramon Obusan** – Promoted ethnic and tribal dances
- **Lucrecia Urtula** – Bayanihan Dance choreographer

Philippine Satellites Through the Years

- 1987 – Agila-1
- First **Philippine satellite** through **private acquisition**
 - Acquired from Indonesia by Mabuhay Satellite Corporation
- 1997 – Agila-2
- First **Philippine-owned satellite launched into space**
 - Launched from **China**
- 2016 – Diwata-1
- First **Philippine-designed and built microsatellite**
 - Deployed from the **International Space Station (ISS)**
 - Built by Filipino engineers under DOST
- 2018 – Maya-1
- First **Philippine-designed and built nanosatellite**
 - Also deployed from the **ISS**
 - Part of the Birds Program in Japan
- 2018 – Diwata-2
- Second **Filipino-built microsatellite**
 - Successor of Diwata-1
 - Also designed to monitor the environment and disaster response
- 2021 – Maya-2, Maya-3, and Maya-4
- Replacements of Maya-1
 - **First batch of university-built satellites** by Filipino students
 - Launched under the BIRDS-2 and BIRDS-4 project with JAXA
- 2023 – Maya-5 and Maya-6
- **Second batch** of university-built satellites
 - Continues the local development of nanosatellite technology

Important Notes

- **Agila-1** – First satellite of the Philippines (acquired)
- **Diwata-1** – First **microsatellite** built and designed by **Filipinos**

- **Maya-1** – First **nanosatellite** built and designed by **Filipinos**
- **Philippine Space Agency (PhilSA)** was established in **2019** but early satellites were developed by **DOST** and partner institutions

Tropical Cyclone Warning System

- **Tropical Depression** – ≤61 kph
- **Tropical Storm** – 62–88 kph
- **Severe Tropical Storm** – 89–117 kph
- **Typhoon** – 118–184 kph
- **Super Typhoon** – ≥185 kph

Life And Works Of Rizal

Jose Protacio Rizal Mercado y Alonzo Realonda

Birth and Death: June 19, 1861 – December 30, 1896

Rizal Law (Republic Act No. 1425)

Pangkalahatang Impormasyon

- **Pangalan ng Batas:** Rizal Law
- **Batas Bilang:** Republic Act No. 1425
- **Petsa ng Pagpasa:** May 17, 1956
- **Petsa ng Paglagda:** June 12, 1956
- **Nilagdaan ni:** Pangulong **Ramon Magsaysay**
- **Pangunahing may-akda:** **Sen. Claro M. Recto**
- **Katuwang na may-akda:** **Sen. Jose P. Laurel Sr.**
- **Layunin:**
 - Ituro sa mga paaralan ang buhay, mga akda, at paniniwala ni **Gat Jose Rizal**, lalo na ang **Noli Me Tangere** at **El Filibusterismo**.
 - Itanim sa puso't isipan ng kabataan ang **nasyonalismo**, **pagkamakabayan**, at **pagkilala sa kasaysayan**.

Kontrobersyal at Oposisyon

Tumutol sa Batas:

- **Simbahang Katolika** at mga grupong panrelihiyon:
 - **Catholic Bishops Conference of the Philippines (CBCP)**
 - **Holy Name Society of the Philippines**
 - **Catholic Action of the Philippines**
 - **Legion of Mary**
 - **Knights of Columbus**
 - **Daughters of Isabela**
- Mga senador tulad nina:
 - **Sen. Francisco Rodrigo**
 - **Sen. Mariano Cuenco**
 - **Sen. Decoroso Rosales**
- Mga kongresistang tulad nina:
 - **Ramon Durano, Marciano Lim, Jose Nuguid, Manuel Soza, Miguel Cuenco, Lucas Paredes, Carmen Consing**, at **Tecia San Andres Ziga**

Batayan ng Pagtutol:

- **Nilalabag daw ang kalayaan sa relihiyon at konsensya**

- Mahigit **170 bahagi ng Noli** at **50 bahagi ng Fili** ang umano'y tumutuligsa sa Simbahang Katolika
- Sinabing si Rizal ay lumabag sa **Canon Law 1399**
- Inakusahan si Sen. Recto ng pagiging komunista

Mga Sumusuporta sa Batas

- **Sen. Claro Recto**
 - "This is not a fight against Recto but a fight against Rizal."
- **Sen. Jose Laurel Sr.**
 - Si Rizal ay tagapagtaguyod ng nasyonalismo
- Iba pang senador:
 - **Sen. Lorenzo Tanedo, Quintin Paredes, Domacao Alonto**
- Mga kongresistang tulad nina:
 - **Jacobo Gonzales, Emilio Cortez, Joaquin Roxas, Lancap Lagumbay, Pedro Lopez**
- **Mayor Arsenio Lacson (Maynila)**
 - Nag-walkout sa misa dahil sa pagtutol ng Simbahan
- Mga grupong pro-Rizal bill:
 - **Knights of Rizal, Women Writers of the Vernacular, Philippine Veterans Legion, College Editor's Guild, Philippine School Teachers Association**

Amendments sa Batas (Compromise)

- Hindi na **sapilitan** ang pagbabasa ng **"unexpurgated" Noli** at **Fili** para sa elementarya at high school
- Maaari nang **humingi ng exemption** ang mga estudyante sa pamamagitan ng affidavit
- Hindi sila exempted sa kurso kay Rizal kahit exempted sa pagbasa ng nobela

Pagpapatupad ng Batas

- **1994:** Inilabas ni **Pres. Fidel V. Ramos** ang **Memorandum Order No. 247**
 - Inatasan ang DepEd at CHED na **buong ipatupad ang Rizal Law**
- **CHED Memorandum No. 3:** Nagpatibay sa implementasyon ng batas sa kolehiyo

Family and Early Life

- **Parents:**
 - Father: **Francisco Mercado Rizal**
 - Mother: **Teodoro Alonso Realonda**
- **Siblings**
 - **Eleven** children, **two boys** and **nine girls**.

1. Saturnina (1850-1913)

- **Oldest** of the Rizal children
- Nickname: **Neneng**
- Married **Manuel T. Hidalgo** of Tanawan, Batangas

2. Paciano (1851-1930)

- Older brother and **confidant** of Jose Rizal
- Was a **second father** to Rizal
- Immortalized in Rizal's novel "Noli Me Tangere" as the wise **Pilosopo Tasio**
- Became a **combat general** in the Philippine Revolution
- Died on April 13, 1930, an old bachelor aged **79**
- Had **two children** by his mistress (*Severina Decena*), a boy and a girl

3. Narcisa (1852-1939)

- Nickname: **Sisa**
- Married Antonio Lopez (nephew of Father Leoncio Lopez), a school teacher of Morong

4. Olympia (1855-1887)

- Nickname: **Ypia**
- Married Silvestre Ubaldo, a telegraph operator from Manila

5. Lucia (1857-1919)

- Married Mariano Herbosa of Calamba
- Herbosa died of cholera in 1889 and was denied Christian burial because he was a brother-in-law of Dr. Rizal

6. Maria (1859-1945)

- Nickname: **Biang**
- Married Daniel Faustino Cruz of Biñan, Laguna

7. Jose (1861-1896)

- The greatest Filipino hero and peerless genius
- Nickname: **Pepe**
- Nickname that his brother and sisters used to call him: **Ute**
- Lived with **Josephine Bracken**, an Irish girl from Hong Kong
- **Had a son** who **died** a few hours after birth; Rizal named him **Francisco** after his father and buried him in **Dapitan**

8. Concepcion (1862-1865)

- Nickname: **Concha**
- **Died** of **sickness** at the age of **3**
- Her death was **Rizal's first sorrow** in life

9. Josefa (1865-1945)

- Nickname: **Panggoy**
- Died an **old maid** at the age of 80

10. Trinidad (1868-1951)

- Nickname: **Trining**
- Died an **old maid** in 1951, aged 83

11. Soledad (1870-1929)

- Nickname: **Choleng**
- **Youngest** of the Rizal children
- Married Pantaleon Quintero of Calamba

- **Surname:** Mercado (real surname)
- **Paternal Great Grandfather:** Domingo Lamco

- **Paternal Grandfather:** Juan Mercado
- **Baptized:** June 22, 1861, by Padre Collantes (ninong/godfather)
- **Early Education:** Learned alphabet (Cartilla) at age 3; could read and write by age 5; started formal education by age 7
- **Baptism Place:** Antipolo Shrine (Virgin of Antipolo) with Don Francisco
- **Physical Traits:** Had a big head, was frail and often sick as a child
- **Pets and Companions:** Usman (big black dog), Alipato (pony)
- **Nurse Maid:** Aya
- **Favorite Pastimes:** Flying kites (Guryon)

Personal Details and Interests

- **Music:** Not strong in music
- **Favorite Fruits:** Lanzones and mango
- **Last Meal:** 3 hard-boiled eggs

Important People

- **Taviel de Andrade:** Lawyer
- **Felipe Luiz Castillo:** Took Rizal's vital signs and pulse rate before execution
- **Siblings and Relatives:**
 - Narcisa "Sisa" (marked Rizal's grave)
 - Mariano (Lucia's husband, died during Cholera Epidemic)
 - Josefa "Punggoy" (had epilepsy)
 - Trinidad "Trining" (last farewell given by Rizal)
 - Soledad (most educated among the sisters)

Historical Context: GOMBURZA

- Mariano Gomez (priest, Bacoar)
- Jacinto Zamora (priest, Marikina)
- Jose Burgos (priest, Manila)

Rizal's Major Works

Noli Me Tangere (1887, Berlin)

Characters:

- **Crisostomo Ibarra:** Filipino educated in Europe for 7 years (Ilustrado), son of Don Rafael Ibarra
- **Maria Clara:** Beautiful daughter of Capitan Tiago (not biological daughter)
- **Elias:** Mysterious friend of Ibarra and master boater who helps him escape
- **Padre Damaso:** Arrogant Franciscan friar
- **Padre Salvi:** Friar in love with Maria Clara; responsible for the death of Crispin (a sacristan boy)
- **Capitan Tiago:** Wealthy businessman and aristocrat, opium addict
- **Sisa:** Devoted mother who went mad looking for her sons Crispin and Basilio
- **Victorina:** Wife of Don Tiburcio de Espadaña (fake doctor), a socialite pretending to be Spanish

El Filibusterismo (1891, Ghent, Belgium)

- **Characters:**
- **Simoun:** Crisostomo Ibarra in disguise as a wealthy jeweler seeking revenge
- **Basilio:** Medical student and assistant to Simoun in his plans
- **Isagani:** Poet and friend of Basilio, emotional and reactive
- **Juli:** Resident of San Diego
- **Don Tiburcio:** Henpecked husband of Victorina, quack doctor
- **Ben Zayb:** Spanish journalist critical of Filipinos
- **Padre Camorra:** Corrupt parish priest who molests Juli
- **Padre Irene:** Head of Spanish Academy, kind
- **Padre Florentino:** Filipino priest who lives by the sea and helps Simoun
- **Plot Points:** Simoun eventually drinks poison and confesses to Padre Florentino

Rizal Biographies and References

- **Palma:** *Biografia de Rizal*
- **Quirino:** *The Great Malayan*
- **Osias:** *Life and Times*
- **Zaide:** *Life, Works, and Writings*
- Rizal is recognized as the **First Filipino Nationalist**

Other Notes

- Maria Clara: Rizal's sweetheart and love interest
- Father Damaso and Pia Alba: Biological parents of Maria Clara
- Plot of *Noli Me Tangere* includes the planned revolution and abduction of Maria Clara
- Maria Clara was raised by Cablesang Tales (Matanglawin), the father of one of the key revolutionaries

Philippine Historical Figures and Currency

Old Philippine Currency (Rab Q Ormas)

- **1 Piso** – Jose Rizal
- **5 Piso** – Emilio Aguinaldo
- **10 Piso** – Bonifacio & Mabini
- **20 Piso** – Manuel L. Quezon
- **50 Piso** – Sergio Osmeña
- **100 Piso** – Manuel Roxas
- **200 Piso** – Diosdado Macapagal
- **500 Piso** – Benigno & Corazon Aquino
- **1000 Piso** – Santos, Escoda, Lim (S.E.L.)

New Generation Currency Series

Coins:

- **1 Piso** – Jose Rizal
- **5 Piso** – Andres Bonifacio
- **10 Piso** – Apolinario Mabini
- **20 Piso (Coin)** – Manuel L. Quezon

Polymer Banknotes:

- **50 Piso** – Visayan Leopard Cat, Taal Lake
- **100 Piso** – Palawan Peacock-Pheasant, Mayon Volcano
- **500 Piso** – Visayan Spotted Deer, Puerto Princesa River
- **1000 Piso** – Philippine Eagle, Tubbataha Reefs

Festivals And Locations

- **Cebu** – Sinulog
- **Kalibo** – Ati-Atihan
- **Iloilo** – Dinagyang

Historical Traitors

- **Macabebe Scouts**
- **Aguinaldo** (controversial figure in some contexts)
- **Galut**
- **Tirad Pass** (associated with Gregorio "Goyo" del Pilar)
- **Makapili** (Japanese collaborators during the Occupation)
- **Patiño**
- **Mariano Gil** (priest who reported the Katipunan)
- **Tirona** (insulted Bonifacio)

Psychology And Learning Theorists

- **Thorndike** (Law of Effect) – Cat experiments
- **Pavlov** – Dog experiments (Classical Conditioning)
- **Skinner** – Rat experiments (Operant Conditioning)
- **Bandura** – Bobo Doll experiment (Social Learning Theory)
- **Little Albert – John B. Watson** (Classical Conditioning)

Vocabulary Words

- **Equanimity** – Calmness and composure
- **Apocalyptic** – Prophetic about the end of the world
- **Sophisticated** – Advanced or complex
- **Sophomoric** – Immature
- **Verbose** – Talkative or using too many words
- **Pulchritude** – Beauty, loveliness
- **Docile** – Easy to manage or teach
- **Amorphous** – Shapeless
- **Hiatus** – Pause or gap
- **Lapse** – Temporary failure
- **Blissful** – Extremely happy
- **Ecstatic** – Overwhelmingly joyful

Philosophers And Sociology

- **David Hume** – No lasting self
- **Karl Marx** – Class struggle in society
- **Auguste Comte** – Sociology as a scientific study of society

Computers Pioneers

- **Charles Babbage** – Father of Computers
- **Alan Turing** – Father of Modern Computing
- **Ada Lovelace** – Mother of Computer Programming

Cultural And Historical Notes

- **Pan de Regla** – First Filipino bread
- **Colado** – Awit (a traditional song, noted as unsold in eateries)
- **Maragtas** – Visayan epic about good manners
- **Bahasa Malaysia** – Asian language closest to Filipino

Random Facts

- **Scepter** is to orb (symbols of royalty)
- **Colosseum** – Not a natural wonder
- **Vladimir Putin** – Leader of Russia (Special Military)
- **Juan Luna** – Rizal posed for him (artist)
- **Gilbert Ryle** – Philosopher, author of *The Concept of Mind*
- **Quiapo** – Known as “Mabangis na Lungsod” (Fierce City)

1987 Philippine Constitution – Legal Basis of Education (Article XIV)

- **Primary legal basis** of Philippine education
- Promotes **inclusive, accessible, and quality education**
- Supports teacher development, academic freedom, and students’ rights

Important Laws Related To Education & Teaching

- **PD 1006** – Professionalization of Teachers (prior to RA 7836)
- **RA 7836** – *Philippine Teachers Professionalization Act of 1994*
- **RA 9293** – Amends RA 7836; allows **non-education grads with 18 units** to take LET
- **RA 6655** – *Free Public Secondary Education Act*
- **RA 10931** – *Free Tertiary Education Act*
- **RA 6728** – *GASTPE* (Government Assistance to Students and Teachers in Private Education)
- **RA 6713** – *Code of Conduct and Ethical Standards for Public Officials and Employees*
- **RA 7277** – *Magna Carta for Disabled Persons*
- **RA 11650** – Inclusion and Services for Learners with Disabilities
- **RA 9155** – *Governance of Basic Education Act* (DepEd law)
- **RA 7722** – Establishes **CHED**
- **RA 7796** – Establishes **TESDA**
- **RA 7784** – *Center of Excellence for Teacher Education*
- **RA 1425** – *Rizal Law*

- **RA 90210** – Integrate IT in Basic Ed
- **RA 10533** – *Enhanced Basic Education Act (K-12)*
- **RA 7877** – *Anti-Sexual Harassment Act*
- **RA 7610** – *Anti-Child Abuse Law*
- **RA 10627** – *Anti-Bullying Act*
- **RA 8049** – *Anti-Hazing Law*
- **RA 9262** – *Anti-VAWC (Violence Against Women and Children)*
- **RA 10173** – *Data Privacy Act*
- **RA 9994** – *Expanded Senior Citizens Act*
- **Batas Pambansa 232** – *Education Act of 1982*
- **RA 10968 (PQF)** – Philippine Qualifications Framework.
- **DO 73, s. 2012** – KPUP (Knowledge, Process, Understanding, Products).
- **DO 8, s. 2015** – WW-PT-QE (Written Work, Performance Task, Quarterly Exam).
- **DO 70, s. 2012** – Lateral entry/exemption for LPs.
- **CMO Nos. 74–83, s. 2017** – CHED standards.
- **CMO No. 78** – BTLED.
- **CMO No. 75** – BSED.
- **CMO No. 74** – BEED.
- **CFT-SEA** – Competency Framework for Teachers in Southeast Asia.
- **TASE** – Tuning Asia South East

Teacher Licensure And Eligibility

- **PBET** – Old exam for teacher licensure (before LET)
- **LET** – Licensure Exam for Teachers
- **PARA-teacher** – Non-licensed teachers (temporary)
- **PRC passing rate** – 75%
- **License without exam** – under specific conditions via reciprocity
- **Refresher course** – Required if not teaching for 5 years
- **18 units (not EDUC)** – Allowed under RA 9293
- **12 units** – Previously allowed; no longer valid
- **No revocation clause** in some laws
- **First legal basis** – PD 1006
- **Present law** – RA 7836 (amended by RA 9293)

MAGNA CARTA FOR PUBLIC SCHOOL TEACHERS (RA 4670)

- Passed: **June 18, 1966**
- **150 days** maternity leave
- Protects rights, salary, benefits, workload, and academic freedom

Branches Of Government (Constitutional Basis)

Article II – Declaration of Principles (Democratic, Republican, Sovereignty of the people)

Article III – Bill of Rights

- Sec. 3: Right to privacy
- Citizenship: *Jus Soli* (by place), *Jus Sanguinis* (by blood)
- Suffrage: 18 years and above

Article IV – Citizenship

- Natural-born and naturalized
- Dual citizenship allowed

Article V – Suffrage

- 18 years old and above
- Registered voter
- Residency requirement applies

Article VI – Legislative Department

- **House of Representatives:** 25 years old, 3-year term, 3 terms
- **Senators:** 35 years old, 6-year term, 2 terms
- **Roles:** Make/amend laws, repeal, enact national budget

Article VII – Executive Department

- **President:**
 - 40 years old, natural-born, 10-year residency
 - **6 years, 1 term**, no reelection
 - Roles: Head of State, Chief Executive, Commander-in-Chief
 - Powers: Pardon, reprieve, commutation, veto bills
- **Vice President:**
 - 40 years old, same qualifications
 - Succeeds President if needed
 - Member of the Cabinet

Impeachable Officials

1. President
2. Vice President
3. Supreme Court Justices
4. Ombudsman
5. Members of Constitutional Commissions

Article VIII – Judicial Department

- **Supreme Court:** 1 Chief Justice, 14 Associate Justices
- **Role:** Interpret laws, settle disputes
- **Retirement:** 70 years old

Article IX – Constitutional Commissions

- A. **Commission on Audit (COA)**
- B. **Commission on Elections (COMELEC)**
- C. **Civil Service Commission (CSC)** – Promotes efficiency, morale, integrity

Sustainable Development Goals (SDGs)

The **17 Sustainable Development Goals (SDGs)** are part of the United Nations' **Agenda 2030**, aiming to end poverty, protect the planet, and ensure peace and prosperity for all.

All 17 SDGs:

1. **No Poverty**
2. **Zero Hunger**
3. **Good Health and Well-being**
4. **Quality Education**
5. **Gender Equality**

6. **Clean Water and Sanitation**
7. **Affordable and Clean Energy**
8. **Decent Work and Economic Growth**
9. **Industry, Innovation, and Infrastructure**
10. **Reduced Inequalities**
11. **Sustainable Cities and Communities**
12. **Responsible Consumption and Production**
13. **Climate Action**
14. **Life Below Water**
15. **Life on Land**
16. **Peace, Justice, and Strong Institutions**
17. **Partnerships for the Goals**

21st Century Learning And Its Classroom Implementation

21st-century learning focuses on **skills needed in modern education and global society**, known as the **4 Cs**:

- **Critical Thinking**
- **Creativity**
- **Collaboration**
- **Communication**

Implementation In The Classroom

- Use of technology (digital tools, e-learning)
- Project-based learning
- Student-centered strategies
- Real-world problem solving
- Emphasis on global awareness and sustainability

Extended Maslow's Hierarchy Of Needs (Revised by Education Experts)

Maslow's original 5 levels were extended to **8 stages**:

1. **Physiological Needs** – food, water, shelter
2. **Safety Needs** – security, protection
3. **Love and Belongingness** – relationships, friends
4. **Esteem Needs** – self-respect, recognition
5. **Cognitive Needs** – knowledge, meaning, self-awareness
6. **Aesthetic Needs** – beauty, balance, form
7. **Self-Actualization** – realizing personal potential
8. **Transcendence** – helping others achieve self-actualization

Mildred Parten Newhall's 6 Stages Of Play (Social Development)

Mildred Parten (1932) identified 6 stages of play in children:

1. **Unoccupied Play** – observing, not playing
2. **Solitary Play** – playing alone (age 0–2)
3. **Onlooker Play** – watching others play
4. **Parallel Play** – playing side by side with others but no interaction
5. **Associative Play** – interacting but not coordinating
6. **Cooperative Play** – organized play with roles and goals (age 4+)

Types Of Learning Difficulties / Disorders

1. **Dyslexia** – difficulty with reading and recognizing basic words
2. **Dysgraphia** – difficulty with writing and spelling
3. **Dyscalculia** – difficulty with numbers and math concepts
4. **Dysphasia / Aphasia** – difficulty in language expression and understanding
5. **Dyspraxia** – difficulty with motor coordination (motor learning disorder)
6. **Dyspepsia** – *not a learning disability* (means indigestion)
7. **Schizophrenia** – mental disorder involving loss of contact with reality
8. **Autism Spectrum Disorder (ASD)** – challenges in social interaction, communication, and behavior
9. **ADHD (Attention Deficit Hyperactivity Disorder)** – problems with attention, focus, impulsivity
10. **Myopia** – nearsightedness
11. **Hyperopia** – farsightedness

The Philippine Qualifications Framework (PQF)

A national policy that **describes the levels of educational qualifications** and sets the **standards for quality assurance** in education and training.

PQF Levels (Level 1 to Level 8):

1. Basic entry-level work
2. Semi-skilled work
3. Skilled work
4. Supervisory work
5. Baccalaureate degree graduate
6. Post-baccalaureate (e.g., Master’s)
7. Advanced research and academic work (Doctorate level)
8. Highly specialized leadership or expertise (e.g., national/international policy-making)

Newborn Reflexes (*Primitive reflexes present at birth; help assess neurological development*)

1. **Rooting Reflex** – baby turns head toward touch on cheek
2. **Sucking Reflex** – baby sucks when mouth is touched
3. **Moro Reflex (Startle Reflex)** – throws arms and legs outward when startled
4. **Palmar Grasp** – baby grasps object when palm is touched
5. **Plantar Grasp** – toes curl when sole of foot is touched
6. **Babinski Reflex** – toes fan out when the sole of the foot is stroked
7. **Tonic Neck Reflex (Fencer’s Position)** – arm extends on the side the head is turned
8. **Stepping Reflex** – baby appears to take steps when feet touch surface
9. **Galant Reflex** – baby curves toward the side when back is stroked

Pen Names / Aliases Of Writers

- **George Orwell** – Blair
- **Mark Twain** – Samuel Clemens
- **Boz** – Charles Dickens
- **Lewis Carroll** – Charles Dodgson
- **George Eliot** – Mary Evans
- **J.K. Rowling** – Robert Galbraith
- **Silence Dogood** – Benjamin Franklin
- **Dr. Seuss** – Theodor Geisel
- **J.D. Salinger** – Jerome David Salinger
- **Harper Lee** – Nelle Harper Lee
- **F. Scott Fitzgerald** – Real name
- **H.G. Wells** – Herbert George Wells
- **J.R.R. Tolkien** – John Ronald Reuel Tolkien
- **Acton Bell, Currer Bell** – Charlotte & Anne Brontë
- **Homer** – Iliad, Odyssey (no confirmed real name)

Filipino Writers & Pen Names

- **Francisco Balagtas** – Balagtas
- **Jose Rizal** – Dimasalang, Laong Laan, Agno
- **Marcelo H. del Pilar** – Plaridel, Piping Dilat, Siling Labuyo, Dolores Manapat
- **Graciano Lopez Jaena** – Diego Laura
- **Andres Bonifacio** – Agapito Bagumbayan, Maypagasa (Magdiwang)
- **Emilio Aguinaldo** – Magdalo
- **Emilio Jacinto** – Pingkian, Dimasilaw
- **Antonio Luna** – Taga-ilog
- **Juan Luna** – Buan
- **Apolinario Mabini** – Bini, Paralitiko, Katabay
- **Jose Palma** – Esteban, Anahaw, Gan Hantik
- **Pascual Poblete** – Anak Bayan
- **Mariano Ponce** – Naning, Tikbalang, Kalipulako
- **Lope K. Santos** – Doktor Lukas, Anak Bayan
- **Severino Reyes** – Lola Basyang
- **Huseng Batute** – Jose Corazon de Jesus
- **Huseng Sisiw** – Jose Dela Cruz
- **Julio Abril, Ernani, Armarite** – Amado Hernandez
- **Jesus Balmori** – Batikuling
- **Guillermo Tolentino** – Fluvio Gil
- **Teodoro A. Agoncillo** – T. A. A.
- **Rio Alma** – Virgilio Almario
- **Kuntil Butil** – Florentino Collantes
- **Kintin Kulirat** – Pepe Peña
- **Catulo, Calipso, Calypso** – Carlos Apostol
- **Quijano de Manila** – Nick Joaquin

Philosophical ISMS

- **Existentialism** – Freedom, individual responsibility
- **Constructivism** – Learning through experience
- **Social Reconstructionism** – Reform society through education

- **Progressivism** – Learn by doing
- **Pragmatism** – Value of usefulness
- **Essentialism** – Core subjects and basics
- **Perennialism** – Timeless truths and classics
- **Idealism** – Reality is mental, spiritual
- **Realism** – Reality exists independent of mind
- **Empiricism** – Knowledge through senses
- **Utilitarianism** – Greatest good for the greatest number
- **Hedonism** – Pursuit of pleasure
- **Stoicism** – Endurance, emotional control
- **Epicureanism** – Pleasure through modesty, wisdom
- **Behaviorism** – Learning via conditioning
- **Humanism** – Human values, dignity

- **Father of English Literature** – Geoffrey Chaucer
- **Father of English Tragedy** – Christopher Marlowe
- **Father of Italian Literature** – Dante Alighieri
- **Father of Ilocano Literature** – Pedro Bukaneg
- **Prince of Filipino Printers** – Tomas Pinpin
- **Goddess of Philippine Poetry** – Leona Florentino
- **First Asian Nobel Laureate for Literature** – Rabindranath Tagore
- **First Filipino Novel in English** – *A Child of Sorrow* by Zoilo Galang
- **First Filipino Novel in Spanish** – *Ninay* by Pedro Paterno

Famous Works / Authors

- **"Pride and Prejudice"** – Jane Austen
- **"A Tale of Two Cities"** – Charles Dickens
- **"A Farewell to Arms", "The Sun Also Rises", "The Old Man and the Sea"** – Ernest Hemingway
- **"Hamlet", "Macbeth", "Othello"** – William Shakespeare
- **"Divine Comedy"** – Dante Alighieri
- **"Paradise Lost"** – John Milton
- **"The Rape of the Lock"** – Alexander Pope
- **"Song of Hiawatha" / "A Psalm of Life"** – Henry Wadsworth Longfellow
- **"War and Peace"** – Leo Tolstoy
- **"Dead Stars"** – Paz Marquez Benitez
- **"Scent of Apples"** – Bienvenido Santos
- **"The Tracks of Babylon"** – Edilberto Tiempo
- **"Alice's Adventures in Wonderland"** – Lewis Carroll
- **"Little Women"** – Louisa May Alcott
- **"How Do I Love Thee"** – Elizabeth Barrett Browning
- **"A Child of Sorrow"** – Zoilo Galang (first English novel in the Philippines)
- **"Ninay"** – Pedro Paterno (first Spanish novel by a Filipino)

LITERARY FORMS & TERMS

- **Epic** – Long narrative (e.g., *Iliad*, *Odyssey*)
- **Tragedy** – Aeschylus, Sophocles, Shakespeare
- **Allegory** – Extended metaphor
- **Fable** – Moral story using animals
- **Myth** – Traditional, often supernatural
- **Legend** – Semi-historical stories
- **Free Verse** – No fixed meter
- **Haiku** – 5-7-5, nature-themed
- **Ode** – Lyric poem expressing praise
- **Sonnet** – 14 lines, iambic pentameter

Literary Titles / Honors

- **The Bard / Bard of Avon** – William Shakespeare
- **Father of English Essay** – Francis Bacon

Fundamental Learning Theories And Their Pioneers

- IVAN PAVLOV – Classical Conditioning**
 - Learning through association
 - Famous dog experiment: salivation at sound of bell
 - Key Concepts:
 - Extinction – learned response fades
 - Discrimination – response to specific stimuli only
 - Generalization – similar stimuli trigger same response
- BURRHUS FREDERIC (B.F.) SKINNER – Operant Conditioning**
 - Behavior shaped by reinforcement or punishment
 - Positive Reinforcement – add reward to increase behavior
 - Negative Reinforcement – remove unpleasant stimulus to increase behavior
 - Punishment – decreases behavior
 - Famous rat experiment
- JOHN B. WATSON – Behaviorism**
 - Known as Father of Behaviorism
 - Believed behavior is learned through conditioning
 - Little Albert experiment (fear conditioning)
 - Promoted Tabula Rasa (blank slate) idea
- JOHN LOCKE – Tabula Rasa (Blank Slate)**
 - Mind is empty at birth
 - Knowledge comes from experience
- EDWARD THORNDIKE – Laws of Learning / Connectionism**
 - Law of Effect – behaviors with positive outcomes are repeated
 - Law of Exercise – practice strengthens learning
 - Law of Readiness – learning happens when learner is ready
 - Learning through trial and error

- Responses are connected to stimuli
- 6. ALBERT BANDURA – Social Learning Theory**
- Learning through observation and imitation
 - Famous Bobo Doll experiment
 - Introduced Self-Efficacy – belief in one’s ability to succeed
- 7. JEROME BRUNER – Modes of Representation**
- Enactive – learning through actions (e.g., play)
 - Iconic – learning through images
 - Symbolic – learning through language/symbols (abstract)
- 8. NOAM CHOMSKY – Language Acquisition Device (LAD)**
- Language learning is innate
 - Children are prewired to learn language
 - Emphasized universal grammar
- 9. URIE BRONFENBRENNER – Ecological Systems Theory**
- Microsystem – immediate surroundings (family, school)
 - Mesosystem – connections between microsystems
 - Exosystem – external influences (e.g., parent’s job)
 - Macrosystem – culture, beliefs, laws
 - Chronosystem – life changes over time
- 10. ROGER SPERRY – Brain Lateralization**
- **Left Brain** – logic, math, language
 - **Right Brain** – creativity, arts, intuition
- 11. JOHN FLAVELL – Metacognition**
- “Thinking about thinking”
 - Ability to monitor and control thinking
 - Three Types of Metacognitive Knowledge:
 - Person Variables – knowing how people think differently
 - Task Variables – knowing what needs to be done
 - Strategy Variables – knowing how to approach a task
- 12. HOWARD GARDNER – Multiple Intelligences**
1. Linguistic – word smart
 2. Logical-Mathematical – number/reasoning smart
 3. Spatial – picture smart
 4. Bodily-Kinesthetic – body smart
 5. Musical – sound smart
 6. Interpersonal – people smart
 7. Intrapersonal – self smart
 8. Naturalistic – nature smart

4 Personality Traits (Temperaments)
1. Sanguine – Extrovert, active, social 2. Choleric – Independent, goal-oriented 3. Melancholic – Analytical, introvert 4. Phlegmatic – Relaxed, peaceful, quiet
Generations
• Silent Generation (1928–1945) – Known for traditional values and discipline • Baby Boomers (1946–1964) – Born after World War II, experienced economic growth • Generation X (1965–1980) – Rise of personal computing, independent and resourceful • Millennials (Gen Y) (1981–1996) – Tech-savvy, value experiences and connectivity • Generation Z (1997–2012) – Digital natives, socially aware, prefer visual communication • Generation Alpha (2013–present) – Growing up fully in digital world, highly connected and tech-dependent
Philippine Festivals & Folk Dances
• Pahiyas Festival – Lucban, Quezon; thanksgiving for a bountiful harvest • Itik-itik – Surigao; dance mimicking duck movements • Tinikling – Leyte; dance imitating tikling birds dodging bamboo traps • Singkil – Maranao of Lake Lanao; royal Muslim dance with intricate steps • Maglalatik – Laguna; war dance depicting battle over coconut latik
Cultural & Historical Sites
• Paoay Church – Ilocos Norte; famous Baroque architecture and earthquake-resistant design • Galápagos Islands – Natural World Heritage Site, known for unique biodiversity
Important Concepts
• Moral – Relates to a person’s values and ethics • Biological – Pertains to a human being’s physical and natural traits • Economic – Refers to productivity and contribution to society • Global Community – The concept of nationalism and unity within diversity
Notable Art, Literature & Mythology
• Epic of Gilgamesh – One of the earliest known literary works • Bakunawa – Mythical moon-eating dragon in Philippine folklore • Venus de Milo – Ancient Greek statue, famous for missing arms • Letras y Figuras – Art technique by Juan de los Reyes Lozano where letters form shapes

- **Winnie the Pooh** – Early 20th century beloved children’s character inspired by stuffed animals
- **Carlos “Botong” Francisco** – National Artist famous for murals like *Bayanihan sa Bukid*

Natural & Cultural Landmarks

- **Banaue Rice Terraces** – Sophisticated, man-made agricultural terraces in the Philippines
- **Rice Terraces** – Often used as examples of ecological harmony with environment

Educational Philosophies

Student-Centered Approaches

- **Existentialism**
 - Emphasizes freedom of choice and individual responsibility
 - Key thinker: Søren Kierkegaard
- **Constructivism**
 - Learning based on prior knowledge and active construction of understanding
 - Key thinker: Lev Vygotsky
- **Social Reconstructionism**
 - Education as a tool to address social injustices and reform society
 - Key thinker: Paulo Freire
 - Examples: ROTC, NSTP programs
- **Progressivism**
 - Focus on teaching skills to adapt to change
 - Key thinkers: John Dewey, Parker
 - Emphasizes multiple intelligences and experiential learning
- **Pragmatism**
 - Learning by doing; practical consequences guide learning
 - Key thinkers: Dewey, Pierce, James
- **Naturalism**
 - Emphasis on natural development and learning through experience
 - Father of Naturalism: Jean-Jacques Rousseau
 - Pestalozzi’s 3Hs: Head, Heart, Hands
 - Survival of the fittest: Herbert Spencer
- **Humanism**
 - Focus on living life to the fullest and personal growth
- **Utilitarianism**
 - Greatest good for the greatest number

Teacher-Centered Approaches

- **Essentialism**
 - Emphasizes teaching the 3Rs (Reading, Writing, Arithmetic) and basic moral conduct
 - Key thinker: William Bagley

- **Behaviorism**
 - Learning shaped by the environment through conditioning
 - Key thinker: John Watson
- **Idealism**
 - Focus on moral and intellectual values
 - Key thinker: Plato
- **Realism**
 - Focus on objective reality, independent of the learner
 - Key thinker: Aristotle
- **Perennialism**
 - Focus on enduring ideas and universal truths
 - Emphasis on broad, general knowledge applicable "then, now, and tomorrow"
- **Empiricism**
 - Knowledge through sensory experience
 - Key thinkers: Hutchins and Adler
- **Rationalism**
 - Knowledge through reasoning and logic

Historical Education Periods In The Philippines

Pre-Spanish Period

- Education informal, survival-focused
- Tribal tutors
- Methods: show and tell, imitation

Spanish Period

- Education used to promote Christianity
- Emphasis on memorization and dictation

First Philippine Republic

- Malolos Constitution (January 21, 1899)
- President: Emilio Aguinaldo
- Promoted love of country and God but less religious influence
- Lacked formal pedagogy
- Free and compulsory elementary education

American Period

- Promotion of democracy and use of English language
- Formal public school system established
- Emphasis on debate and games

Commonwealth Period

- 1935 Constitution (May 14, 1935)
- Leaders: Manuel Quezon, Sergio Osmeña
- Preparation for independence
- Promotion of nationalism
- Use of Filipino language

Japanese Occupation

- 1943 Constitution
- President: Jose P. Laurel
- Education focused on reason, logic (Descartes) and love of labor (tech-vocational skills)
- Used Nippongo (Japanese language)
- Promoted eradication of dependence on Westerners

Third Philippine Republic

- Used 1935 Constitution
 - Presidents: Roxas, Quirino, Magsaysay, Garcia, Macapagal
 - Emphasized social and economic development
- Martial Law Era / Marcos Sr. Administration**
- 1973 Constitution (January 17, 1979)
 - Presidential Proclamation 1102
 - Department of Education and Culture (1972) changed to Ministry of Education and Culture (1978)
 - Created 13 regional offices
 - Focus on qualitative and quantitative growth in education
- Post-EDSA Period**
- 1987 Constitution (February 2, 1987)
 - President: Corazon Aquino onwards
 - Department reorganizations:
 - MECS: Ministry of Education, Culture, and Sports (1982)
 - DECS: Department of Education and Sports (1987)

Career Stages (BPHD)

- **Beginning** – Level 6
- **Proficient** – Independent
- **Highly Proficient** – Supports and mentors colleagues
- **Distinguished** – Has a global perspective on teaching and learning

Power Of Teachers

- **Expert Power** – Mastery of subject and pedagogy
- **Referent Power** – Influence through consensus and respect
- **Reward Power** – Uses grades and incentives
- **Coercive Power** – Uses authority and consequences

Parenting Styles

- **Authoritative** – High expectations with high support (Democratic)
- **Authoritarian** – High expectations with no support (Strict, aggressive)
- **Permissive** – No expectations, but warm and loving (Spoils the child)
- **Neglectful** – No expectations, no support (Uninvolved)

Technology Integration Matrix

- **Entry** – Beginning use of technology
- **Adoption** – Procedural use (step-by-step)
- **Adaptation** – Exploratory use (trying different tech)
- **Infusion** – Learners choose the tools
- **Transformation** – Learning becomes impossible without technology

Transfer Of Learning

- **Positive Transfer** – Previous learning helps new learning
- **Negative Transfer** – Previous learning interferes with new learning
- **Zero Transfer** – Previous learning has no effect on new learning

Teacher Standards

- **NCBTS** – *National Competency-Based Teachers Standards*
- **PPST** – *Philippine Professional Standards for Teachers*
- **SBM** – *School-Based Management*

Kinds Of Leave

- **Vacation Leave** – 15 days per year
- **Sick Leave** – 15 days per year
- **Maternity Leave** – 105 days fully paid (up to 4 children); half-pay starting with the 5th child
- **Paternity Leave** – 7 days fully paid
- **Study Leave** – 1 year with 60% of monthly salary
- **Overtime Pay** – 25% of monthly salary
- **Hazard Pay** – For risky working environments

Motivation

- **Intrinsic Motivation** – Doing something because of genuine interest (e.g., "*Gusto mo talaga*")
- **Extrinsic Motivation** – Doing something for external rewards (e.g., medals, grades)

Preventive Approach To Classroom Management

- **Planned Ignoring** – Deliberately ignoring minor misbehavior
- **Proximity Control** – Standing near the student (Lapitan)
- **Physical Restraint** – Changing seat to prevent disruption
- **Direct Appeal** – Verbally telling the student to stop ("*Tumigil ka!*")
- **Signal Interference** – Using signals (e.g., "shhh") to get attention
- **Removal of Object** – Taking away distracting items

Basic Linguistics Concepts

- **Phonology** – Study of sounds
- **Syntax** – Sentence structure
- **Semantics** – Study of meaning (Tip: "*seMEANtics*")
- **Morphology** – Study of word formation and structure

Political Thinkers

- **Benigno "Ninoy" Aquino** – *Political Idealist*
- **Julius Caesar** – *Political Realist*

Types Of Rights

- **Disenfranchise** – Deprived of the right to vote
- **Civil Rights** – Rights that promote individual happiness and freedoms
- **Political Rights** – Right to run for public office
- **Suffrage** – Right to vote
- **Natural Rights** – Rights connected to nature and human dignity
- **Statutory Rights** – Rights granted by law (e.g., extended maternity leave)
- **Constitutional Rights** – Rights protected by the Constitution (e.g., Bill of Rights)

Writs

- **Writ of Amparo** – Protection of the right to life, liberty, and security
- **Writ of Habeas Corpus** – "Bring the body"; protects against illegal detention
- **Writ of Habeas Data** – Protects information privacy
- **Writ of Kalikasan** – Protects environmental rights

Types Of Taxes

- **Income Tax** – Paid by professionals and employees
- **Community Tax** – Commonly known as *Cedula*
- **Tariff** – Tax on imported goods
- **Ad Valorem Tax** – Tax based on value, applied to luxury goods
- **Sin Tax** – Imposed on harmful products (e.g., imported liquor)
- **Corporate Tax** – Tax paid by businesses
- **Real Estate Tax** – Tax on properties
- **Inheritance Tax** – Paid by the heirs
- **Capital Gains Tax** – Paid by property sellers

Inherent Powers Of The State

- **Police Power** – Ensures general welfare
- **Eminent Domain (Expropriation)** – Taking private property for public use (e.g., road widening)
- **Taxation** – Government's power to impose financial burdens for public purposes

Trade Laws During The American Period

- **Payne-Aldrich Act** – Partial free trade
- **Underwood-Simmons Act** – Full free trade between the Philippines and the U.S.

Factors Of Production (CELL)

- **Capital** – Money
- **Entrepreneur** – Skill in managing business
- **Land** – Natural resources
- **Labor** – Human effort

Elements Of A State (mnemonic: PSG Te)

- **Population** – People
- **Sovereignty** – Supreme power or authority
- **Government** – System of rule
- **Territory** – Land area

Filipina Heroines And National Women

- **Joan of Arc (Philippine Counterparts):**
 - **Gregoria Montoya** – From Cavite
 - **Teresa Magbanua** – From Visayas
 - **Agueda Kahabagan** – From Tagalog region
- **Marcela Agoncillo** – Sewed the Philippine Flag
- **Trinidad Tecson** – "Mother of the Philippine Red Cross"
- **Josefa Llanes Escoda (Guerrero)** – Helped lepers and WWII prisoners (also known as the *Leperspy*)
- **Gabriela Silang** – Wife of Diego Silang; revolutionary leader
- **Gregoria de Jesus** – Wife of Andres Bonifacio; "Lakambini of Katipunan"
- **Corazon Aquino** – "Mother of Philippine Democracy"
- **Melchora Aquino** – "Mother of the Katipunan"
- **Sisa** – Fictional character; represents abused women
- **Cariñosa** – Traditional Philippine folk dance

West Philippine Sea Dispute

Claimants:

1. **Taiwan**
2. **Vietnam**
3. **Malaysia**
4. **China**
5. **Brunei**

Famous Philosophers And Their Views On The Self

1. SOCRATES

- *"Know thyself"*
- *"The unexamined life is not worth living"*
- *"I know that I don't know"*
- Believed in **Dualism** – the body and the soul are distinct

2. PLATO

- *"Necessity is the mother of invention."*
- Student of Socrates
- Believed in the existence of the **Mind** and **Immortal Soul**
- Also supported **Dualism**:
 - **Appetitive** – basic physical needs
 - **Spirited** – emotions
 - **Rational** – reason and intellect

3. ARISTOTLE

- Student of Plato
- Believed the **soul is the essence of the self**
- Rejected dualism; the **body and soul are one**

4. ST. AUGUSTINE

- "Our heart is restless until it finds rest in You."
- Happiness is the **ultimate end goal** of man
- Without God, humans are **doomed to fail**

5. RENE DESCARTES

- Known as the **Father of Modern Philosophy**
- *Cogito ergo sum* – "I think, therefore I am."
- Believed in **Mind (Cognito)** and **Body (Extensa)**

6. SIGMUND FREUD

- Founder of **Psychoanalysis**
- The self is composed of both **conscious and unconscious** parts
- **Structural Model of the Psyche:**
 - **Id** – instinctual drives
 - **Ego** – reality, balance
 - **Superego** – morals and conscience

7. GILBERT RYLE

- "The self is the way people behave."
- Rejected the idea of a separate soul
- Self is **behavioral** – what a person does

8. PAUL CHURCHLAND

- Proponent of **Eliminative Materialism**
- "The self is defined by the activities of the brain."
- "The self is nothing but the brain."

9. MAURICE MERLEAU-PONTY

- Combined **Existentialism** and **Phenomenology**
- Believed in the **embodied self:**
 - **Subjective body** – lived experience
 - **Objective body** – physical/observable body

10. JOHN LOCKE

- Known as the **Father of Classical Liberalism**
- Personal identity = **consciousness**
- Introduced **Tabula Rasa** – the mind as a "blank slate"
- Knowledge is gained **through experience**

11. DAVID HUME (corrected from "Humid")

- Scottish philosopher
- Advocated **Empiricism, Skepticism, and Naturalism**
- Denied the existence of a permanent self
- "What we call the mind is nothing but a bundle of perceptions."

12. IMMANUEL KANT

- Bridged **Empiricism and Rationalism**
- Introduced the **Transcendental Unity of Apperception**
- The self is shaped by the **apparatus of the mind** – mental structures that process experience

Number Sense And Number Theory

Type of Number	Description	Example
Natural Numbers	Counting numbers	1, 2, 3, 4, ...
Whole Numbers	Natural numbers + zero	0, 1, 2, 3, ...
Integers	Whole numbers + negatives	...-2, -1, 0, 1, 2...
Rational Numbers	Can be expressed as a fraction (includes integers, decimals, repeating decimals)	½, -3, 0.75, 2.333...
Irrational Numbers	Cannot be written as a fraction; non-repeating, non-terminating	√2, π, e
Real Numbers	All rational and irrational numbers	All above
Imaginary Numbers	Involve the square root of a negative number	√-1 = i
Complex Numbers	Combination of real and imaginary	3 + 2i

Place Value System

- **Ones, Tens, Hundreds, Thousands, etc.**
- **Decimal places:** Tenths, Hundredths, Thousandths
- **Read numbers correctly:**
 - Example: 3,205.67 = *Three thousand two hundred five and sixty-seven hundredths*

Number Comparisons

- Use <, >, = signs
- Know **ascending (least to greatest)** and **descending (greatest to least)** order
- Convert to same format first when comparing fractions and decimals

Factors and Multiples

- **Factor:** Number that divides another without remainder
 - Example: Factors of 12 → 1, 2, 3, 4, 6, 12
- **Multiple:** Product of a number and any whole number
 - Example: Multiples of 4 → 4, 8, 12, 16...

Prime and Composite Numbers

- **Prime Number:** Only divisible by 1 and itself
 - Examples: 2, 3, 5, 7, 11, 13 (2 is the only even prime!)
- **Composite Number:** More than two factors
 - Examples: 4, 6, 8, 9, 10

Greatest Common Factor (GCF)

- Largest number that divides two or more numbers
- **Method:** List factors or use prime factorization

- Example: GCF of 12 and 18 = 6
- Least Common Multiple (LCM)**
- Smallest number that is a multiple of two or more numbers
 - **Method:** List multiples or use prime factorization
 - Example: LCM of 4 and 6 = 12

Divisibility Rules	
Rule for	If...
2	Last digit is even
3	Sum of digits divisible by 3
4	Last two digits divisible by 4
5	Ends in 0 or 5
6	Divisible by 2 and 3
9	Sum of digits divisible by 9
10	Ends in 0

- Prime Factorization**
- Break down number into **prime factors**
 - Use **factor tree**
 - Example: $60 \rightarrow 2 \times 2 \times 3 \times 5$ or $2^2 \times 3 \times 5$

- Rounding & Estimation**
- Round to nearest 10, 100, etc.
 - Use in checking answer reasonableness
 - Ex: Round 368 to nearest hundred = 400

- Even vs. Odd**
- **Even:** Ends in 0, 2, 4, 6, 8
 - **Odd:** Ends in 1, 3, 5, 7, 9

Roman Numerals	
Symbol	Value
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

- Set Notations**
- Union: $A \cup B$
 - Intersection: $A \cap B$
 - Complement: A'
 - Subset: $A \subseteq B$
 - Empty Set: $\emptyset$
 - Universal Set: ε

Operations And Properties Of Numbers

Basic Operations

- 1. Addition (+)**
- Combining numbers to get a total.
 - Properties:
 - **Commutative:** $a + b = b + a$
 - **Associative:** $(a + b) + c = a + (b + c)$
- 2. Subtraction (−)**
- Finding the difference between numbers.
 - *Not* commutative or associative.
 - Always subtract in order: $a - b \neq b - a$
- 3. Multiplication (×)**
- Repeated addition of the same number.
 - Properties:
 - **Commutative:** $a \times b = b \times a$
 - **Associative:** $(a \times b) \times c = a \times (b \times c)$
 - **Distributive:** $a(b + c) = ab + ac$
- 4. Division (÷)**
- Splitting a number into equal parts.
 - Division is *not* commutative: $a \div b \neq b \div a$
 - Can be expressed as fractions: $a \div b = a/b$

- Order of Operations (PEMDAS Rule)**
- To evaluate expressions correctly, follow this order:
1. **P** – Parentheses
 2. **E** – Exponents (or roots)
 3. **MD** – Multiplication and Division (left to right)
 4. **AS** – Addition and Subtraction (left to right)

Properties of Real Numbers		
Property Name	Description	Example
Commutative	Order doesn't matter (only + or ×)	$a + b = b + a$
Associative	Grouping doesn't matter (only + or ×)	$(a + b) + c = a + (b + c)$
Distributive	Multiplying over addition/subtraction	$a(b + c) = ab + ac$
Identity	Keeps value the same	$a + 0 = a, a \times 1 = a$
Inverse	Gets back to the identity	$a + (-a) = 0, a \times (1/a) = 1$
Zero Property	Any number $\times 0 = 0$	$5 \times 0 = 0$

- Operations with Fractions & Decimals**
- **Addition/Subtraction:** Use common denominator
 - **Multiplication:** Multiply numerators & denominators directly
 - **Division:** Multiply by the reciprocal of the divisor

Example:
 $(2/3) \div (4/5) = (2/3) \times (5/4) = 10/12 = 5/6$

- Conversions and Applications**
- **Percent to Decimal:** Divide by 100
 - $25\% \rightarrow 0.25$
 - **Decimal to Fraction:** Read the place value
 - $0.75 \rightarrow 75/100 = 3/4$

- **Fraction to Percent:** Multiply by 100
 - $\frac{2}{5} \times 100 = 40\%$

Algebra (Basics)

Algebraic Expressions

- **Definition:** A combination of numbers, variables, and operations.
→ Example: $3x + 2$, $5a^2 - 7a + 4$
- **Terms:** Each part of the expression separated by + or –
- **Coefficient:** The number multiplied by the variable
→ In $5x$, the coefficient is 5.
- **Constant:** A term without a variable
→ In $x + 4$, the constant is 4.

Simplifying Expressions

- Combine like terms (same variable and exponent)
 - Example: $3x + 4x = 7x$
- Use **distributive property**:
 - $a(b + c) = ab + ac$
 - Example: $2(x + 3) = 2x + 6$

Solving Linear Equations

- Goal: Isolate the variable.
- Steps:
 1. Simplify both sides.
 2. Move variables to one side.
 3. Move constants to the other.
 4. Solve for the unknown.

Example:

$$\begin{aligned} 2x + 3 &= 11 \\ \rightarrow 2x &= 11 - 3 \\ \rightarrow 2x &= 8 \\ \rightarrow x &= 4 \end{aligned}$$

Solving Inequalities

- Same steps as equations.
- When multiplying/dividing both sides by a **negative**, reverse the inequality symbol.

Example:

$$\begin{aligned} -2x &> 6 \\ \rightarrow x &< -3 \end{aligned}$$

Symbols:

- $>$: greater than
- $<$: less than
- $\geq$: greater than or equal
- $\leq$: less than or equal

Translating Word Problems to Algebra

Common keywords:

- "is" $\rightarrow =$
- "more than" $\rightarrow +$
- "less than" $\rightarrow -$
- "times" $\rightarrow \times$
- "divided by" $\rightarrow \div$

Linear Equations and Graphing

- **Standard Form:** $Ax + By = C$
- **Slope-Intercept Form:** $y = mx + b$
 - m = slope (rise/run)

- b = y-intercept (point where line crosses y-axis)

Systems of Linear Equations

- Solving for **two variables** using:
 - Substitution method
 - Elimination method
 - Graphing method

Quadratic Equations (Basic)

- **Standard Form:** $ax^2 + bx + c = 0$
- **Factoring method:**
 - Example: $x^2 - 5x + 6 = 0$
→ $(x - 2)(x - 3) = 0$
→ $x = 2$ or $x = 3$
- **Other Methods:** Completing the square, quadratic formula

Sequences and Series

- **Arithmetic Sequence:**
 - Formula: $a_n = a_1 + (n - 1)d$
 - Constant difference
 - Example: 2, 5, 8, 11, ...
- **Geometric Sequence:**
 - Formula: $a_n = a_1 \times r^{n-1}$
 - Constant ratio
 - Example: 3, 6, 12, 24, ...

Geometry

Basic Geometry Terms

- **Point** – exact location, no size or shape
- **Line** – straight path with no thickness, extends forever
- **Line segment** – part of a line with two endpoints
- **Ray** – starts at one point and extends endlessly in one direction
- **Plane** – flat surface extending in all directions

Types of Angles

- **Acute:** less than 90°
- **Right:** exactly 90°
- **Obtuse:** between 90° and 180°
- **Straight:** exactly 180°

Triangle Properties

- **Sum of angles** = 180°
- **Types by sides:**
 - Equilateral – 3 equal sides
 - Isosceles – 2 equal sides
 - Scalene – no equal sides
- **Types by angles:**
 - Acute – all angles $< 90^\circ$
 - Right – one 90° angle
 - Obtuse – one angle $> 90^\circ$

Common Triangle Formulas:

- **Area** = $\frac{1}{2} \times \text{base} \times \text{height}$
- **Pythagorean Theorem** (Right triangle):
 - $a^2 + b^2 = c^2$, where c = hypotenuse

- **Heron’s Formula** (when all sides are known):

$$A = \sqrt{s(s - a)(s - b)(s - c)}$$

where:

$$s = \frac{a + b + c}{2}$$

- A = area of the triangle
- a, b, c = lengths of the sides
- s = semi-perimeter (half of the triangle's perimeter)

Quadrilaterals

- **Sum of angles** = 360°
- **Types:**
 - **Square:** all sides and angles equal
 - **Rectangle:** opposite sides equal, all angles = 90°
 - **Parallelogram:** opposite sides equal and parallel
 - **Rhombus:** all sides equal, opposite angles equal
 - **Trapezoid:** one pair of parallel sides

Common Formulas:

- **Area of rectangle** = length × width
- **Area of square** = side²
- **Area of trapezoid** = ½ × (base₁ + base₂) × height

Circles

- **Parts:**
 - Radius – center to edge
 - Diameter – 2 × radius
 - Circumference – distance around the circle
- **Formulas:**
 - **Area** = $\pi \times r^2$
 - **Circumference** = $2\pi r$ or πd
 - **Arc length** = $(\theta/360) \times 2\pi r$
 - **Sector area** = $(\theta/360) \times \pi r^2$ (where θ is the central angle)

Solid Geometry (3D Shapes)

Includes Volume and Surface Area of solids

Shape	Volume	Surface Area
Cube	$V = s^3$	$SA = 6s^2$
Rect. Prism	$V = l \times w \times h$	$SA = 2(lw + lh + wh)$
Cylinder	$V = \pi r^2h$	$SA = 2\pi r^2 + 2\pi rh$
Cone	$V = (1/3)\pi r^2h$	$SA = \pi r^2 + \pi rl$
Sphere	$V = (4/3)\pi r^3$	$SA = 4\pi r^2$
Pyramid	$V = (1/3) \times \text{Base Area} \times \text{height}$	$SA = \text{Base Area} + \text{Lateral Area}$

Perimeter & Area

Shape	Perimeter	Area
Square	4s	s ²
Rectangle	2(l + w)	lw
Triangle	sum of 3 sides	½bh
Circle	2πr or πd	πr ²
Parallelogram	2(l + w)	bh
Trapezium	a + b + c + d	½(a + b)h
Rhombus	4s	½d ₁ d ₂

Surface Area & Volume

Solid	Surface Area	Volume
Cuboid	2(lw + lh + wh)	lwh
Cylinder	2πrh + 2πr ²	πr ² h
Cone	πr√(r ² + h ²) + πr ²	⅓πr ² h
Sphere	4πr ²	⅔πr ³

POLYGONS

Sum of Interior Angles

- Formula: $(n - 2) \times 180^\circ$

Polygon	Interior Sum
Pentagon	540°
Hexagon	720°
Heptagon	900°
Octagon	1080°
Nonagon	1260°
Decagon	1440°
Icosagon	3240°

Coordinate Geometry

- **Slope** of a line:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- **Distance Formula:**

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- **Midpoint Formula:**

$$m = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Equation of a Line:**

- $y = mx + b$ (slope-intercept form)

Statistics

Measures of Central Tendency

1. Mean (Average)

- For Ungrouped Data (Raw Scores)

$$\text{Mean} = \frac{\sum x}{n}$$

- Σx = Sum of all values
- n = Number of values

- For Grouped Data (Frequency Table)

$$\text{Mean} = \frac{\sum (f \cdot x)}{\sum f}$$

- f = Frequency
- x = Midpoint of each class
- $\Sigma (f \cdot x)$ = Sum of frequency $\times$ midpoint
- Σf = Total frequency

2. Median

- Middle value when data is ordered.
- For Ungrouped Data

► If n is odd:

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ value}$$

► If n is even:

$$\text{Median} = \frac{\left(\frac{n}{2} \right)^{\text{th}} \text{ value} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ value}}{2}$$

- For Grouped Data

$$\text{Median} = L + \left(\frac{\frac{n}{2} - F}{f} \right) \times h$$

Note: Always arrange data in ascending order before applying.

3. Mode

- Most frequent value in the dataset.
- A set can be:
 - Unimodal (1 mode)
 - Bimodal (2 modes)
 - Multimodal (3+ modes)
 - No mode (no repeated values)
- For Ungrouped Data

Mode = Most frequent value

- For Grouped Data

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{(2f_1 - f_0 - f_2)} \right) \times h$$

- L = Lower boundary of the modal class
- f_1 = Frequency of the modal class
- f_0 = Frequency before the modal class
- f_2 = Frequency after the modal class
- h = Class width

Comparison Tip:

- **Mean** is sensitive to outliers.
- **Median** is best when data has extreme values.
- **Mode** is useful for categorical data (e.g., favorite color).

Measures of Dispersion (Spread)

1. Range

- Formula:

$$\text{Range} = \text{Highest value} - \text{Lowest value}$$

2. Variance (σ^2 or s^2)

- Measures how far numbers are spread out from the mean.
- Higher variance = more spread.

Population Variance (σ^2)

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Sample Variance (s^2)

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

- x = each data point
- μ = population mean
- $\bar{x}$ = sample mean
- N = population size
- n = sample size

3. Standard Deviation (σ or s)

- Square root of the variance.
- Shows average distance from the mean.
- The **smaller** the SD, the more **consistent** the data.

Population Standard Deviation (σ)

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Sample Standard Deviation (s)

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Frequency Distribution

- **Table** showing how often values occur.
- **Class Interval:** Grouped data
- **Frequency:** Number of values per class
- May include:

- Cumulative Frequency
- Relative Frequency (%)

Graphical Presentation of Data

1. **Bar Graph** – Comparing categories
2. **Histogram** – For continuous data (no gaps)
3. **Pie Chart** – Shows proportions (360° total)
4. **Line Graph** – Trends over time
5. **Ogive** – Cumulative frequency graph

Data Types

- **Quantitative**: Measurable (height, age)
- **Qualitative**: Categories (gender, color)
- **Discrete**: Countable (no decimals)
- **Continuous**: Any value within a range

Quartiles, Deciles, Percentiles

- **Quartiles (Q1, Q2, Q3)**: Divide data into 4 parts
- **Deciles (D1–D9)**: Divide data into 10 parts
- **Percentiles (P1–P99)**: Divide into 100 parts
- **Use**: Locate position of a score in a dataset
 - *Example*: 75th percentile = score is higher than 75% of data

Probability Connection

- Statistics is often linked with **Probability**
- Know how to:
 - Interpret outcomes using statistics
 - Calculate likelihood based on data

Probability

Probability is the measure of how likely an event is to happen.

Formula:

- **Range of Probability**:
 - 0 = Impossible event
 - 1 = Certain event
 - All probabilities lie between 0 and 1

Key Concepts:

1. **Experiment** – An activity that produces outcomes (e.g., tossing a coin, rolling a die).
2. **Sample Space (S)** – The set of all possible outcomes.
 - *Example*: Tossing a coin $\rightarrow S = \{H, T\}$
3. **Event (E)** – A specific outcome or group of outcomes.
 - *Example*: Getting heads when tossing a coin

Examples of Common Probability Scenarios

- **Tossing a Coin**
 - $S = \{H, T\}$
 - $P(H) = \frac{1}{2}$
- **Rolling a Die (1 to 6)**
 - $S = \{1, 2, 3, 4, 5, 6\}$
 - $P(\text{rolling a 4}) = \frac{1}{6}$
 - $P(\text{even number}) = \frac{3}{6} = \frac{1}{2}$
- **Drawing from a Deck of 52 Cards**
 - Red cards = 26; Black cards = 26
 - $P(\text{Heart}) = \frac{13}{52} = \frac{1}{4}$
 - $P(\text{Jack or Queen}) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$

Types of Events

● Independent Events

Events that do not affect each other.

- *Example*: Tossing a coin and rolling a die

● Dependent Events

Events that affect each other.

- *Example*: Drawing 2 cards without replacement

● Mutually Exclusive Events

Events that cannot happen at the same time.

- *Example*: Getting heads and tails in one coin toss

● Complementary Events

The sum of the probabilities of an event and its complement is 1.

Helpful Hints for Solving Probability in LEPT

- Use a **table** or **list** to organize sample space.
- Watch out for **"with"** or **"without"** replacement:
 - *With replacement*: total outcomes stay the same.
 - *Without replacement*: total outcomes decrease.
- Read the question carefully: sometimes they ask for **at least**, **at most**, or **neither**.
- Always **reduce fractions** to the lowest terms.

Problem Solving & Word Problems

General Strategy for Solving Word Problems (Polya's 4 Steps)

1. Understand the Problem

- Read the problem carefully.
- Identify **what is being asked**.
- Highlight/underline **important information**.
- Watch for keywords (e.g., "total," "difference," "each").

2. Devise a Plan

- Choose a strategy:
 - Use a formula
 - Draw a diagram
 - Make a table
 - Guess and check
 - Use logical reasoning
 - Work backward
 - Create an equation

3. Carry Out the Plan

- Perform calculations step-by-step.
- Keep units consistent (e.g., hours vs. minutes).
- Double-check math operations.

4. Check and Interpret the Answer

- Does the answer make sense?
- Reread the problem and verify if the answer satisfies the conditions.

Common Types of Word Problems

Problem Type	Clue Words / Tips	Example Keywords
Addition	Combine, total, altogether, sum	"How many in all?"
Subtraction	Difference, left, fewer, less than	"How many more?"
Multiplication	Times, product, total groups	"Each," "per," "every"
Division	Split equally, shared, quotient	"How many each?"
Comparison	More than, fewer than, as much as	"Compare," "difference"
Age Problems	Present, past, future ages	Use variables like x
Money Problems	Profit, loss, cost, selling price	"How much earned?"
Distance-Speed-Time	$D = R \times T$	Use correct units
Work Problems	Work = Rate $\times$ Time	Combine rates for groups
Mixture Problems	Use percentage $\times$ quantity	"What is the strength?"
Ratio & Proportion	Use cross-multiplication	"Find the missing part"

SAMPLE WORD PROBLEM STRATEGIES

1. Use Algebra (Let statements)

Problem: Ana is twice as old as Bea. The sum of their ages is 36. What are their ages?

Let: Bea = x $\rightarrow$ Ana = 2x

Equation: $x + 2x = 36$

Solve: $3x = 36 \rightarrow x = 12 \rightarrow$ Ana = 24

2. Use Tables or Charts for Organizing Data

Especially for time, rate, or inventory problems.

3. Work Backward

Problem: A number is added to 5, then multiplied by 2, and the result is 24. What is the number?

Work backwards:

$24 \div 2 = 12 \rightarrow 12 - 5 = 7$

Measurement Conversion Table

Length

From	To	Conversion Factor
1 meter (m)	centimetres (cm)	100 cm = 1 m
1 cm	millimeters (mm)	10 mm = 1 cm
1 kilometer (km)	meters (m)	1,000 m = 1 km
1 inch (in)	centimeters (cm)	2.54 cm = 1 in
1 foot (ft)	inches (in)	12 in = 1 ft
1 yard (yd)	feet (ft)	3 ft = 1 yd
1 mile (mi)	feet (ft)	5,280 ft = 1 mi

Weight (Mass)

From	To	Conversion Factor
1 kilogram (kg)	grams (g)	1,000 g = 1 kg
1 gram (g)	milligrams (mg)	1,000 mg = 1 g
1 pound (lb)	ounces (oz)	16 oz = 1 lb
1 kilogram (kg)	pounds (lb)	2.2046 lb = 1 kg

Capacity (Volume)

From	To	Conversion Factor
1 liter (L)	milliliters (mL)	1,000 mL = 1 L
1 gallon (gal)	quarts (qt)	4 qt = 1 gal
1 quart (qt)	pints (pt)	2 pt = 1 qt
1 pint (pt)	cups	2 cups = 1 pt
1 gallon (gal)	liters (L)	3.785 L = 1 gal

Time

From	To	Conversion Factor
1 minute (min)	seconds (s)	60 s = 1 min
1 hour (hr)	minutes (min)	60 min = 1 hr
1 day	hours (hr)	24 hr = 1 day
1 week	days	7 days = 1 week
1 year	days	365 days = 1 year

Temperature

Formula to convert	Description	Formula to convert
$^{\circ}\text{F} = (^{\circ}\text{C} \times 1.8) + 32$	Celsius to Fahrenheit	$^{\circ}\text{F} = (^{\circ}\text{C} \times 1.8) + 32$
$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \div 1.8$	Fahrenheit to Celsius	$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \div 1.8$

Business Math Basics

Percentage

- Part out of 100.
- To change percent to decimal: divide by 100 (e.g., 25% = 0.25).
- To change decimal to percent: multiply by 100 (e.g., 0.6 = 60%).
- Formula:
Percentage = (Part ÷ Whole) × 100

Discounts

- Money taken off the original price.
- Discount amount = Original price × Discount rate
- Final price = Original price – Discount amount

Commission

- Money earned as a percent of sales.
- Commission = Sales × Commission rate

Tax

- Extra money added to price.
- Tax amount = Price × Tax rate
- Total price = Price + Tax amount

Simple Interest

- Extra money earned or paid over time on a loan or savings.
- Interest = Principal × Rate × Time
- Total amount = Principal + Interest

Profit and Loss

- Profit = Selling price – Cost price
- Loss = Cost price – Selling price
- Percent profit or loss = (Profit or Loss ÷ Cost price) × 100

Markup

- Money added to cost price to get selling price.
- Selling price = Cost price + Markup

ADDITIONAL REVIEWER: COMMON LEPT QUESTIONS MARCH 2025

- **Neolithic vs. Paleolithic:**
 - *Paleolithic*: Old Stone Age, hunting-gathering, nomadic life.
 - *Neolithic*: New Stone Age, agriculture, settled communities.
- **Father of Economics: Adam Smith**

- **Meaning of Economics:**
 - Study of how people use resources to produce and distribute goods and services.
- **Father of Geography: Eratosthenes**
- **Spoliarium Painter: Juan Luna**
- **Region of Timberland: Valley**
- **Meaning of Geography:**
 - Study of Earth’s physical features, climate, and human-environment interaction.
- **Rice Terraces:** Natural and *not* man-made.
- **Taj Mahal:** *Man-made* (not natural)
- **Essentialism in Education:**
 - Focus on core knowledge: **3Rs** – Reading, 'Riting, 'Rithmetic.
- **Antonyms vs. Synonyms:**
 - *Happy–Sad* is **Antonyms** (opposites).
 - *Synonym* means **similar** (remember: “SAME-Synonym”).
- **Reflect Question:**
 - Requires deeper thought; often asks “why” or “how.”
- **Doble Kara:**
 - Means *double-faced* or *two-faced* (character trait).
- **Applied vs. Pure Research:**
 - *Pure Research*: Knowledge for the sake of understanding.
 - *Applied Research*: Solves practical problems.
- **Puberty vs. Adolescence:**
 - *Puberty* – Physical development (sexual apparatus).
 - *Adolescence* – Broader stage (emotional, cognitive, and social changes).
- **West Valley Fault:**
 - Active fault line in Luzon; earthquake-prone.
- **Ramon Magsaysay:**
 - Known as “**Champion of the Masses.**”
- **Dance from Panay Island: Cariñosa**
- **Dance from the PH known abroad: Tinikling**
- **Human Rights rather than Civil Rights:**
 - *Human Rights*: Universal (life, freedom, dignity)
 - *Civil Rights*: Legal rights within a state (e.g., right to vote)
- **Rizal Timeline (in order)**
 1. Birth
 2. Education
 3. Writing of *Noli Me Tangere*
 4. Writing of *El Filibusterismo*
 5. Arrest, exile, trial
 6. Execution (Dec 30, 1896)

Facebook vs. TikTok (2025)

- **Facebook** remains the **most used** platform worldwide.
- **Facebook users (2025): 3.07 billion**
- **TikTok users (2025): 1.59 billion**
- **TikTok** is more popular with **Gen Z**, but **Facebook** has **wider global reach**
- But in 2022, **Tiktok has the most users than Facebook.**